SIMULTANEOUS RUIN ON BROWNIAN METRIC TREES WITH TERMINAL GROUPS

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Abstract. We derive the asymptotic probability that all leaves of a Brownian metric tree exceed their respective linear barriers at a common time. We cut a fixed finite skeleton at distance κ/u^2 from its terminal height and graft an independent finite genealogy onto each terminal branch. The ruin probability is asymptotic to a Gaussian tail for the skeleton multiplied by a covariance correction and the expected volume of a union of boxes indexed by time. Each box uses one time for all active groups and the union accounts for crossings before and after the terminal branching. For exceedance at the terminal time the corresponding coefficient factorises over the active groups. Both asymptotics include the covariance loss from shortening the skeleton and the Gaussian factor for binding constraints with zero multipliers. Singleton groups recover the fixed-tree result with coordinatewise drifts and shifts. For two leaves with zero drifts and a common barrier we obtain the transition formula at every rate at which the terminal edge length tends to zero, including the correction to the Gaussian exponent. We also derive bounds for the coefficients, identify the active leaves by a finite procedure on the associated resistor network and obtain asymptotics for exceedance by at least one leaf and for the largest distance between leaves.

1. Introduction

Brownian decision trees describe trajectories that share a Brownian history up to a deterministic split and have independent increments on distinct descendant edges thereafter. In a level-synchronous tree all particles in a generation split at the same time into the same number of children. Exact finite-horizon asymptotics are known for exceedance by at least one branch, for the largest distance between branches and for simultaneous exceedance by all branches [2]. We study simultaneous ruin when several groups of leaves branch inside the terminal excursion window. A fixed finite Brownian tree supplies the skeleton. From time $T-\kappa/u^2$ we continue leaf j by a fixed finite genealogy $\mathcal{G}_j$ whose times and spatial increments are scaled by u^{-2} and u^{-1}, respectively. Denote its descendant set by $\mathcal{B}_j$ and the resulting paths by $X_{ji}^{(u)}$, $i\in\mathcal{B}_j$. All descendants in group j have the common barrier $q_ju+c_jt-x_j$ whereas these parameters may differ between groups. The main probability is

$$\Psi_{\mathcal{B}}(u)=\mathbb{P}\Big\{\exists t\in[0,T]: X_{ji}^{(u)}(t)-c_jt+x_j>q_ju,\ j=1,\dots,d,\ i\in\mathcal{B}_j\Big\}.$$

The skeleton, horizon, scaled genealogies and barrier parameters are fixed in the first main theorem. The second main theorem treats two leaves with an arbitrary shrinking terminal length, including nonconvergent scaled lengths $u^2\delta(u)$.

Theorem 2.1 identifies one conditioning offset for each skeleton leaf with a positive quadratic-programming multiplier. That offset is shared by all of its terminal descendants. The sharp ruin asymptotic is the product of a skeleton Gaussian tail and the expected volume of a union of boxes, with one radial coordinate per active group. Each box uses the same time for all groups. The corresponding terminal coefficient factorises over the active genealogies. Both formulas account for the covariance loss from shortening the skeleton and include the

Date: 1st September 2025.

2020 *Mathematics Subject Classification.* 60G70, 60G15, 60J65.

Key words and phrases. Brownian metric tree, simultaneous ruin, Gaussian extremes, terminal genealogy, quadratic programming.

conditional Gaussian factor for binding constraints with zero multipliers. The ruin coefficient also accounts for exceedance on the reversed trunks before the fine branching.

When every group consists of one leaf, the entire grafted process has the same law as the fixed skeleton. Its terminal coefficient is one, and its ruin coefficient is exactly the fixed-tree Brownian constant. This yields Corollary 2.2 with arbitrary fixed coordinatewise drifts and shifts. The centred fixed-tree case is a specialisation of the theorem for non-stationary vector-valued Gaussian processes (GPs) in [1, Theorem 2.4(2)]. Remark A.9 verifies that identification, including possible covariance singularity before the final skeleton branch points. The quadratic-programming reduction of Gaussian orthant tails is also standard; see [3]. The grouped triangular family requires the conditioning, domination and localisation argument given here.

Theorem 2.3 yields the transition for two leaves whose terminal edge length $\delta(u)$ tends to zero. Its Gaussian exponent correction applies at every shrinking rate, even when $u^2\delta(u)$ oscillates. The proof combines conditioning at the split with terminal Gaussian tail estimates for the degenerating covariance. The one-group finite-genealogy formula appears in Example 3.2; the Appendix verifies it independently by reflection.

Proposition 3.4 and Lemma 3.5 identify the skeleton quadratic cost with a network obstacle problem and provide a finite procedure for finding its active leaves. The examples show unequal currents, active-set and boundary transitions, and explicit trend effects. Example 3.3 treats two sister pairs, with a diagram, a factorised terminal coefficient and a two-dimensional ruin constant; the three-leaf and singleton cases are recovered as variations. Proposition 3.1 controls truncation of the reversed trunks for numerical evaluation. The one-branch and largest-distance results extend the corresponding decision-tree formulas in [2].

Our analysis covers fixed finite terminal groups and the transition at all shrinking rates for one two-leaf family. The following extensions require estimates beyond those established here.

(1) *Trends in the general vector-valued setting.* Fixed linear drifts and offsets are included here, with common parameters within each terminal group. For a general deterministic trend g_i, the local quantity to control is

$$u(g_i(T - h_u s) - g_i(T)),$$

where h_u is the endpoint scale. Here $h_u = u^{-2}$, so a fixed linear trend disappears from the local field while its terminal value remains in the Gaussian tail and boundary factor. Other Gaussian models require appropriate conditioning, integrable bounds and localisation. A varying threshold cannot simply be substituted into a fixed-parameter theorem.

(2) *Changing terminal genealogies and local barriers.* The first theorem permits several fixed finite genealogies inside a window of length κ/u^2. Its parameters are fixed as $u \to \infty$; it does not justify substitution of a changing $\kappa(u)$ or changing scaled branching times. The second theorem settles this issue for two leaves and shows why a terminal Gaussian exponent correction can be necessary. For general groups, uniform tail estimates, changing genealogy geometry, and unequal descendant barriers on the local scale remain to be treated. The absence of branching from each bounded excursion window alone does not provide the required uniform estimates.

(3) *Growing trees and changing active sets.* A growing number of leaves or a growing horizon requires control of covariance inverses, small positive multipliers, closing inactive gaps and local constants. The bound $1 \le \mathcal{H}_{\mathcal{I}} \le 2^{|\mathcal{I}|}$ does not supply this uniformity. A boundary residual may enter the local functional when multipliers or inactive gaps vary, instead of remaining an external Gaussian factor. For simultaneous ruin the skeleton quadratic cost and active groups are central. For $d(u)$ leaf exceedance events

$A_i(u)$ the Bonferroni inequalities yield $\mathbb{P}\left\{\bigcup_{i=1}^{d(u)} A_i(u)\right\} \sim \sum_{i=1}^{d(u)} \mathbb{P}\{A_i(u)\}$ whenever

$$\sum_{1\le i<j\le d(u)} \mathbb{P}\{A_i(u)\cap A_j(u)\} = o\left(\sum_{i=1}^{d(u)} \mathbb{P}\{A_i(u)\}\right).$$

Negligibility of each fixed pair does not imply this relation when the number of leaves grows.

(4) *Non-Brownian edge fields.* The exponents governing increment roughness and terminal variance loss need not be the same. For independent copies of fractional Brownian motion (fBM) along the edges, a terminal edge of fixed length ℓ has increment variance h^{2H}, whereas its terminal variance loss is $\ell^{2H} - (\ell - h)^{2H} \sim 2H\ell^{2H-1}h$. Thus the Brownian balance occurs only at $H = 1/2$. The local field, normalisation and localisation must be established in the resulting regimes, including interactions with trends and zero-multiplier constraints.

Section 2 introduces the skeleton and terminal groups, states the two main theorems, and identifies the fixed-tree special case. Section 3 contains the proofs, coefficient bounds and examples, the network interpretation, and complementary exceedance results. Section A contains the auxiliary estimates, the complementary exceedance proofs, independent fixed-tree and one-group verifications, the comparison with the centred vector-valued theorem, and the detailed alternative conditioning proof.

2. Main results

In the first theorem, the finite skeleton, horizon, barrier parameters and scaled terminal genealogies are fixed, while the resulting fine tree varies with u. All limits are as $u \to \infty$. Branching times and offspring numbers may vary from vertex to vertex. The second theorem treats a two-leaf family with an arbitrary shrinking terminal length.

Let $\mathcal{T}$ be the fixed finite rooted skeleton with vertex set $\mathcal{V}$, edge set $\mathcal{E}$, root ρ, and terminal leaves

$$\mathcal{L} = \{1, \dots, d\}.$$

Each vertex v has deterministic height $t(v) \in [0, T]$, with

$$t(\rho) = 0, \qquad t(i) = T, \quad i \in \mathcal{L},$$

and height is strictly increasing along each edge. For an edge $e = (v, w)$, write

$$\ell_e = t(w) - t(v) > 0.$$

We shall assume that each terminal edge has positive length: if $p(i)$ is the parent of leaf i, then

$$T - t(p(i)) > 0, \qquad i = 1, \dots, d.$$

Attach to every edge e an independent standard Brownian motion W_e on $[0, \ell_e]$. For a leaf i, define $B_i(t)$ by summing the edge increments accumulated up to height t on the path from ρ to i. Inside an edge this includes its partial increment. At a vertex it is the sum over the completed edges. Then each B_i is a standard Brownian motion, and the paths of two leaves agree up to the height of their most recent common ancestor.

For $i \neq j$, let $i \wedge j$ be the most recent common ancestor of i and j, and put

$$r_{ij} = t(i \wedge j).$$

With $\boldsymbol{B}(t) = (B_1(t), \dots, B_d(t))^\top$, define

$$\Gamma(t) = \mathrm{Cov}(\boldsymbol{B}(t)).$$

Then

$$\Gamma_{ij}(t) = \begin{cases} t, & i = j, \\ \min(t, r_{ij}), & i \neq j. \end{cases}$$

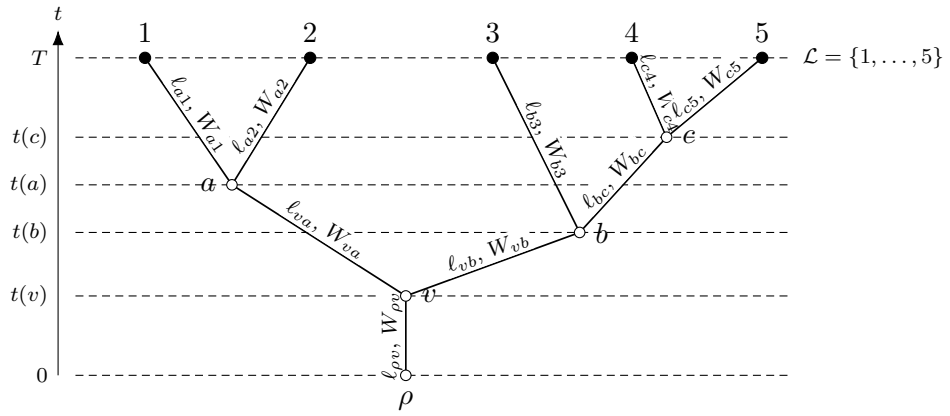


FIGURE 1. Example of a finite rooted Brownian metric tree. Each edge $e = (p, w)$ has length $\ell_e = t(w) - t(p)$ and carries an independent Brownian motion W_e. All skeleton leaves lie at the common terminal height T; the added terminal genealogies are not shown.

In particular, the terminal covariance matrix $\Sigma = \Gamma(T)$ satisfies

$$\Sigma_{ij} = \begin{cases} T, & i = j, \\ r_{ij}, & i \neq j. \end{cases}$$

The terminal-edge assumption implies $\Sigma \succ 0$: every non-trivial linear combination of $B_1(T), \dots, B_d(T)$ contains at least one independent terminal-edge increment with non-zero coefficient.
Let $\mathcal{F}_t$ be the natural filtration generated by the edge Brownian motions up to height t. Then $\boldsymbol{B}$ is a continuous vector-valued GP with independent increments: for $0 \leq s \leq t \leq T$, the increment $\boldsymbol{B}(t) - \boldsymbol{B}(s)$ is independent of $\mathcal{F}_s$, has covariance $\Gamma(t) - \Gamma(s)$, and this covariance matrix has non-negative entries.
Choose

$$0 < \delta_0 < \min_{1 \leq i \leq d} (T - t(p(i))).$$

For $t \in [T - \delta_0, T]$, no branching occurs on $[t, T]$, and hence

$$\Gamma(t) = \Sigma - (T - t) I_d.$$

The edge-Brownian construction also defines a centred Gaussian random field (GRF) $G(x)$, $x \in \mathcal{T}$, on the rooted metric tree. If $x \wedge y$ denotes the highest common ancestor of the two points $x, y \in \mathcal{T}$, then

$$\mathrm{Cov}(G(x), G(y)) = t(x \wedge y).$$

For a leaf i, let $\gamma_i(t)$ be the point of height t on the root-to-i geodesic. Then

$$B_i(t) = G(\gamma_i(t)), \qquad 0 \leq t \leq T.$$

Thus the Brownian metric tree is a particular Gaussian metric tree whose restriction to each root-to-leaf path is Brownian motion. Conditional on the value at a branch point, the fields in its distinct daughter subtrees are independent.
Let

$$\boldsymbol{q} \in (0, \infty)^d, \qquad \boldsymbol{c}, \boldsymbol{x} \in \mathbb{R}^d.$$

The fixed-skeleton reference probability is

$$\Psi_{\mathcal{T}}(u) = \mathbb{P}\{\exists t \in [0, T] : B_i(t) - c_i t + x_i > q_i u, \ i = 1, \dots, d\}.$$

Put

$$\boldsymbol{a}(t) = t\boldsymbol{c} - \boldsymbol{x}, \qquad \boldsymbol{a} = \boldsymbol{a}(T) = T\boldsymbol{c} - \boldsymbol{x},$$

so that the terminal threshold is $u\boldsymbol{q}+\boldsymbol{a}$, and define

$$p_T(u)=\mathbb{P}\{\boldsymbol{B}(T)>u\boldsymbol{q}+\boldsymbol{a}\}.$$

We add terminal groups to the skeleton inside the excursion window. Fix $\kappa>0$. For each skeleton leaf $j\in\mathcal{L}$, let $\mathcal{G}_j$ be a fixed finite rooted metric tree with positive edge lengths, root height zero, and all leaves at height κ. Denote its leaf set by $\mathcal{B}_j$. Independent standard Brownian motions on its edges define trajectories $G_{ji}(s)$, $i\in\mathcal{B}_j$, $0\le s\le\kappa$, by the same construction as above. The genealogies are independent across j and independent of $\boldsymbol{B}$. Put $\mathcal{B}=(\mathcal{B}_1,\dots,\mathcal{B}_d)$. For all sufficiently large u, set

$$r_u=T-\frac{\kappa}{u^2}$$

and define the descendant trajectories

$$X_{ji}^{(u)}(t)=\begin{cases}B_j(t), & 0\le t\le r_u,\\ B_j(r_u)+u^{-1}G_{ji}\big(u^2(t-r_u)\big), & r_u<t\le T,\end{cases}\qquad j\in\mathcal{L},\quad i\in\mathcal{B}_j. \tag{2.1}$$

Thus the edges of $\mathcal{G}_j$ are compressed by u^{-2} in time, while their Brownian increments are scaled by u^{-1}. All descendants of skeleton leaf j inherit its barrier $uq_j+a_j(t)$. Define

$$\Psi_{\mathcal{B}}(u)=\mathbb{P}\Big\{\exists t\in[0,T]:X_{ji}^{(u)}(t)>uq_j+a_j(t),\ j\in\mathcal{L},\ i\in\mathcal{B}_j\Big\}, \tag{2.2}$$

$$p_{\mathcal{B}}(u)=\mathbb{P}\Big\{X_{ji}^{(u)}(T)>uq_j+a_j,\ j\in\mathcal{L},\ i\in\mathcal{B}_j\Big\}. \tag{2.3}$$

The skeleton covariance Σ and the barrier vector $\boldsymbol{q}$ determine the active set. The boundary factor also depends on the terminal shift $\boldsymbol{a}=T\boldsymbol{c}-\boldsymbol{x}$. Each active coordinate represents one terminal group.

Apply Lemma A.1 to $\Pi_\Sigma(\boldsymbol{q})$. Let $\widetilde{\boldsymbol{q}}$ be the minimiser, let $\mathcal{I}$ be the active set, and put

$$\mathcal{J}=\{1,\dots,d\}\setminus\mathcal{I}.$$

Then

$$\widetilde{\boldsymbol{q}}_{\mathcal{I}}=\boldsymbol{q}_{\mathcal{I}},\qquad \widetilde{\boldsymbol{q}}_{\mathcal{J}}=\Sigma_{\mathcal{J}\mathcal{I}}\Sigma_{\mathcal{I}\mathcal{I}}^{-1}\boldsymbol{q}_{\mathcal{I}}\ge\boldsymbol{q}_{\mathcal{J}},$$

with the second identity omitted when $\mathcal{J}=\varnothing$. Moreover

$$\boldsymbol{\lambda}_{\mathcal{I}}:=\Sigma_{\mathcal{I}\mathcal{I}}^{-1}\boldsymbol{q}_{\mathcal{I}}>\boldsymbol{0}_{\mathcal{I}},\qquad \boldsymbol{\lambda}_{\mathcal{J}}=\boldsymbol{0}_{\mathcal{J}},\qquad \boldsymbol{\lambda}=\Sigma^{-1}\widetilde{\boldsymbol{q}}.$$

We use "active set" for the positive-multiplier set $\mathcal{I}$. A constraint can be binding, $\widetilde{q}_j=q_j$, while $j\in\mathcal{J}$ and $\lambda_j=0$. The factor below includes these boundary coordinates. The leading quadratic cost is

$$Q_\Sigma(\boldsymbol{q})=\boldsymbol{q}_{\mathcal{I}}^\top\Sigma_{\mathcal{I}\mathcal{I}}^{-1}\boldsymbol{q}_{\mathcal{I}}=\widetilde{\boldsymbol{q}}^\top\Sigma^{-1}\widetilde{\boldsymbol{q}}.$$

For the inactive coordinates, define $\boldsymbol{b}_{\mathcal{J}}\in[-\infty,\infty)^{|\mathcal{J}|}$ by

$$\boldsymbol{b}_{\mathcal{J}}=\lim_{u\to\infty}\left(u\left(\boldsymbol{q}_{\mathcal{J}}-\Sigma_{\mathcal{J}\mathcal{I}}\Sigma_{\mathcal{I}\mathcal{I}}^{-1}\boldsymbol{q}_{\mathcal{I}}\right)+\boldsymbol{a}_{\mathcal{J}}-\Sigma_{\mathcal{J}\mathcal{I}}\Sigma_{\mathcal{I}\mathcal{I}}^{-1}\boldsymbol{a}_{\mathcal{I}}\right).$$

Equivalently, for $j\in\mathcal{J}$

$$b_j=\begin{cases}a_j-\Sigma_{j\mathcal{I}}\Sigma_{\mathcal{I}\mathcal{I}}^{-1}\boldsymbol{a}_{\mathcal{I}}, & \text{if } q_j=\Sigma_{j\mathcal{I}}\Sigma_{\mathcal{I}\mathcal{I}}^{-1}\boldsymbol{q}_{\mathcal{I}},\\ -\infty, & \text{if } q_j<\Sigma_{j\mathcal{I}}\Sigma_{\mathcal{I}\mathcal{I}}^{-1}\boldsymbol{q}_{\mathcal{I}}.\end{cases}$$

Let $\boldsymbol{Z}\sim N(\boldsymbol{0},\Sigma)$. Set

$$\pi_{\mathcal{J}}(\boldsymbol{b}_{\mathcal{J}})=\mathbb{P}\{\boldsymbol{Z}_{\mathcal{J}}>\boldsymbol{b}_{\mathcal{J}}\mid\boldsymbol{Z}_{\mathcal{I}}=\boldsymbol{0}_{\mathcal{I}}\},$$

with $\pi_{\mathcal{J}}=1$ if $\mathcal{J}=\varnothing$. Coordinates with $b_j=-\infty$ are ignored. Since the conditional Gaussian law is non-degenerate on finite boundary coordinates

$$\pi_{\mathcal{J}}(\boldsymbol{b}_{\mathcal{J}})\in(0,1].$$

For $L>0$, define

$$H_{\mathcal{I}}(L;\boldsymbol{\lambda}_{\mathcal{I}}):=\int_{\mathbb{R}^{|\mathcal{I}|}}\mathbb{P}\big\{\exists s\in[0,L]:\lambda_i W_i(s)-\lambda_i^2 s>z_i,\ i\in\mathcal{I}\big\}\exp\left(\sum_{i\in\mathcal{I}}z_i\right)\mathrm{d}\boldsymbol{z}_{\mathcal{I}},$$

where W_i, $i\in\mathcal{I}$, are independent standard Brownian motions. The map $L\mapsto H_{\mathcal{I}}(L;\boldsymbol{\lambda}_{\mathcal{I}})$ is non-decreasing, and Lemma A.4 yields

$$\mathcal{H}_{\mathcal{I}}(\boldsymbol{\lambda}_{\mathcal{I}}):=\lim_{L\to\infty}H_{\mathcal{I}}(L;\boldsymbol{\lambda}_{\mathcal{I}})\in[1,2^{|\mathcal{I}|}].$$

Let W_j, $j\in\mathcal{I}$, be independent standard Brownian motions, independent of every terminal genealogy. On $K=(-\infty,\kappa]$, define

$$\xi_{ji}(t)=\begin{cases}W_j(-t)-\lambda_j(-t), & t\le 0,\\ G_{ji}(t), & 0\le t\le\kappa,\end{cases}\qquad i\in\mathcal{B}_j. \tag{2.4}$$

All descendants in one group share the same reversed trunk. With

$$m=|\mathcal{I}|,\qquad Q_*=\sum_{j\in\mathcal{I}}\lambda_j^2,\qquad L_j=\min_{i\in\mathcal{B}_j}(G_{ji}(\kappa)),$$

set

$$C_{\mathcal{B}}=\exp(-\kappa Q_*/2)\mathbb{E}\left\{\mathrm{Leb}_m\left(\bigcup_{t\in K}\prod_{j\in\mathcal{I}}\left(0,\exp\left(\lambda_j\min_{i\in\mathcal{B}_j}(\xi_{ji}(t))\right)\right)\right)\right\}, \tag{2.5}$$

$$D_{\mathcal{B}}=\exp(-\kappa Q_*/2)\mathbb{E}\left\{\exp\left(\sum_{j\in\mathcal{I}}\lambda_j L_j\right)\right\}=\exp(-\kappa Q_*/2)\prod_{j\in\mathcal{I}}\mathbb{E}\{\exp(\lambda_j L_j)\}. \tag{2.6}$$

Here Leb_m is m-dimensional Lebesgue measure. The same time t is used in every group in (2.5); the union therefore accounts for simultaneous exceedance. The time interval $(-\infty,0]$ accounts for crossing before the fine branching, and $\exp(-\kappa Q_*/2)$ accounts for the covariance lost by cutting the skeleton at r_u.

Theorem 2.1. *Keep the skeleton, the horizon T, the parameters $(\boldsymbol{q},\boldsymbol{c},\boldsymbol{x})$, $\kappa>0$, and all terminal genealogies fixed. Then $C_{\mathcal{B}}$ and $D_{\mathcal{B}}$ are finite and strictly positive, and, as $u\to\infty$*

$$\Psi_{\mathcal{B}}(u)\sim C_{\mathcal{B}}p_T(u),\qquad p_{\mathcal{B}}(u)\sim D_{\mathcal{B}}p_T(u). \tag{2.7}$$

The common reference tail satisfies

$$\begin{aligned}p_T(u)\sim\pi_{\mathcal{J}}(\boldsymbol{b}_{\mathcal{J}})&\frac{u^{-m}}{(2\pi)^{m/2}|\Sigma_{\mathcal{I}\mathcal{I}}|^{1/2}\prod_{j\in\mathcal{I}}\lambda_j}\\ &\times\exp\left(-\frac{1}{2}(u\boldsymbol{q}_{\mathcal{I}}+\boldsymbol{a}_{\mathcal{I}})^{\top}\Sigma_{\mathcal{I}\mathcal{I}}^{-1}(u\boldsymbol{q}_{\mathcal{I}}+\boldsymbol{a}_{\mathcal{I}})\right).\end{aligned} \tag{2.8}$$

Consequently we have

$$\frac{\Psi_{\mathcal{B}}(u)}{p_{\mathcal{B}}(u)}\longrightarrow\frac{C_{\mathcal{B}}}{D_{\mathcal{B}}}. \tag{2.9}$$

Binding constraints with zero multipliers contribute through $\pi_{\mathcal{J}}(\boldsymbol{b}_{\mathcal{J}})$ in both absolute asymptotics; they do not introduce additional coordinates into (2.5) or (2.6).

If every $\mathcal{G}_j$ has a single leaf, the grafted process has the law of the fixed skeleton for all sufficiently large u, since the cut then lies on the terminal skeleton edges. In this case

$$C_{\mathcal{B}}=\mathcal{H}_{\mathcal{I}}(\boldsymbol{\lambda}_{\mathcal{I}}),\qquad D_{\mathcal{B}}=1, \tag{2.10}$$

for every fixed $\kappa>0$. Thus Theorem 2.1 contains the following fixed-tree result, including its drifts, shifts, and Gaussian boundary factor.

Corollary 2.2. *For the fixed skeleton with positive terminal edge lengths*

$$\Psi_{\mathcal{T}}(u) \sim \mathcal{H}_{\mathcal{I}}(\boldsymbol{\lambda}_{\mathcal{I}})p_T(u),$$

where the full Gaussian asymptotic for $p_T(u)$ is (2.8). Thus singleton groups recover the fixed-tree formula with arbitrary fixed leafwise drifts and shifts.

The second theorem concerns a tree whose terminal edge length depends on u. Write $\overline{\Phi}(x) = \mathbb{P}\{Z > x\}$, where $Z \sim N(0,1)$, and let ϕ denote its density. For independent standard Brownian motions W_1, W_2 and $\kappa \geq 0$, define

$$M_\kappa = \sup_{0\leq s\leq \kappa} \min(W_1(s), W_2(s)), \qquad C_\kappa = 2\exp(-\kappa/2)\mathbb{E}\{\cosh(M_\kappa)\},$$

$$D_\kappa = 2\overline{\Phi}(\sqrt{\kappa/2}), \qquad R_\kappa = C_\kappa/D_\kappa, \qquad \mathcal{H}_2 = \mathcal{H}_{\{1,2\}}(\mathbf{1}_2).$$

These definitions yield $M_0 = 0$, $C_0 = 2$, $D_0 = 1$, and $R_0 = 2$. The finiteness of C_κ follows from $M_\kappa \leq \sup_{0\leq s\leq\kappa} W_1(s)$ and Gaussian exponential moments.

Theorem 2.3. *Let $\delta = \delta(u) \in (0,1)$ satisfy $\delta(u) \to 0$, and put $\kappa = \kappa(u) = u^2\delta(u)$. Let two standard Brownian leaves share their path up to time $1-\delta(u)$ and have independent Brownian increments from the split until time 1. With zero drifts and common barrier u, denote the simultaneous exceedance probability on $[0,1]$ by $\Psi^{\mathrm{sh}}(u)$, and its terminal counterpart by $p^{\mathrm{sh}}(u)$. Define*

$$F(u,\delta) = \left(1 - \frac{\delta}{2}\right)^{3/2} \exp\left(-\frac{u^2\delta^2}{4(2-\delta)}\right).$$

Then, along every such choice of $\delta(u)$

$$\begin{aligned} p^{\mathrm{sh}}(u) &\sim D_{\kappa(u)}\overline{\Phi}(u)F(u,\delta(u)), \\ \Psi^{\mathrm{sh}}(u) &\sim C_{\kappa(u)}\overline{\Phi}(u)F(u,\delta(u)). \end{aligned} \tag{2.11}$$

In particular

$$\Psi^{\mathrm{sh}}(u) \sim R_{\kappa(u)}p^{\mathrm{sh}}(u). \tag{2.12}$$

No convergence assumption on $\kappa(u)$ is required. If $\kappa(u) \to \infty$, these formulas yield

$$p^{\mathrm{sh}}(u) \sim \frac{(2-\delta)^{3/2}}{2\pi u^2\sqrt{\delta}} \exp\left(-\frac{u^2}{2-\delta}\right), \qquad \Psi^{\mathrm{sh}}(u) \sim \mathcal{H}_2 p^{\mathrm{sh}}(u). \tag{2.13}$$

Remark 2.4. *The factor $F(u,\delta)$ accounts for the curvature of the terminal Gaussian exponent. If $\kappa(u) = o(u)$, it tends to one, and the uncorrected expression $C_{\kappa(u)}\overline{\Phi}(u)$ remains valid. If $\kappa(u)/u \to \tau \in [0,\infty)$, then*

$$F(u,\delta(u)) \longrightarrow \exp(-\tau^2/8).$$

For example, $\delta(u) = 1/u$ yields

$$\frac{\Psi^{\mathrm{sh}}(u)}{C_u\overline{\Phi}(u)} \longrightarrow \exp(-1/8).$$

Thus, substituting a growing $\kappa(u)$ into the fixed-κ asymptotic requires control of both the terminal Gaussian tail and the branching window.

3. Further results and proofs

We first prove the grouped theorem and its fixed-tree specialisation. We then derive bounds for the coefficients and examples with one and two terminal groups, followed by the network interpretation of the skeleton and further examples. Finally, we establish the shrinking-edge identities and coefficient limits, prove the all-rates theorem, and state complementary exceedance results. Section A contains the auxiliary estimates, complementary proofs and independent verifications.

Proof of Theorem 2.1. The reference-tail expansion is Lemma A.3. Put $A = \Sigma_{\mathcal{I}\mathcal{I}}$, $m = |\mathcal{I}|$, and let ϕ_A be the centred Gaussian density with covariance A. Write $S_{\mathcal{J}} = \Sigma_{\mathcal{J}\mathcal{J}} - \Sigma_{\mathcal{J}\mathcal{I}} A^{-1} \Sigma_{\mathcal{I}\mathcal{J}}$ for the conditional covariance on the inactive coordinates, omitting it if $\mathcal{J}$ is empty.
Fix $L > 0$, write $h = L + \kappa$, and cut the skeleton at $S_u = T - h/u^2$. Let $M_u(L)$ be the probability of a crossing restricted to $[S_u, T]$. Condition on

$$\boldsymbol{B}_{\mathcal{I}}(S_u) = u\boldsymbol{q}_{\mathcal{I}} + \boldsymbol{a}_{\mathcal{I}}(S_u) - \boldsymbol{z}/u, \qquad \boldsymbol{z} \in \mathbb{R}^m.$$

Its covariance is $A_u = A - hI_m/u^2$, and, for fixed $\boldsymbol{z}$

$$\frac{\phi_{A_u}(u\boldsymbol{q}_{\mathcal{I}} + \boldsymbol{a}_{\mathcal{I}}(S_u) - \boldsymbol{z}/u)}{\phi_A(u\boldsymbol{q}_{\mathcal{I}} + \boldsymbol{a}_{\mathcal{I}})} \longrightarrow \exp(\boldsymbol{\lambda}_{\mathcal{I}}^\top \boldsymbol{z} - hQ_*/2). \tag{3.1}$$

Indeed, $A_u^{-1} = A^{-1} + hu^{-2}A^{-2} + O(u^{-4})$ and $\boldsymbol{a}_{\mathcal{I}}(S_u) - \boldsymbol{a}_{\mathcal{I}} = O(u^{-2})$. The latter shift contributes $o(1)$ to the logarithmic density ratio.
All minima below are over the descendants $i \in \mathcal{B}_j$ of the indicated group. The rescaled active future paths have the law

$$Y_{ji,L}(s) = \begin{cases} U_j(s), & 0 \le s \le L, \\ U_j(L) + G_{ji}(s - L), & L \le s \le h, \end{cases}$$

where U_j are independent standard Brownian motions, independent of the genealogies. The rescaled change in the linear barrier is $c_j s/u$, which is uniformly negligible. Conditional inactive residuals have covariance tending to $S_{\mathcal{J}}$ and thresholds tending to $\boldsymbol{b}_{\mathcal{J}}$. Their future fluctuations, including all finitely many descendants, are $O_{\mathbb{P}}(u^{-1})$ uniformly on $[S_u, T]$. They are independent of the active future paths. Sandwiching the inactive thresholds between $\boldsymbol{b}_{\mathcal{J}} - \eta$ and $\boldsymbol{b}_{\mathcal{J}} + \eta$ on finite coordinates, and then letting $\eta \downarrow 0$, yields the conditional limit

$$\pi_{\mathcal{J}}(\boldsymbol{b}_{\mathcal{J}}) J_L(\boldsymbol{z}), \qquad J_L(\boldsymbol{z}) = \mathbb{P}\{\exists s \in [0, h] : Y_{ji,L}(s) > z_j,\ j \in \mathcal{I},\ i \in \mathcal{B}_j\},$$

for Lebesgue-almost every $\boldsymbol{z}$. Almost-everywhere convergence suffices: these crossing probabilities are coordinatewise monotone functions of $\boldsymbol{z}$, and their discontinuities form a Lebesgue-null set.
For completeness, fix $0 < \varepsilon < \min_{j \in \mathcal{I}}(\lambda_j)$. For large u the ratio in (3.1) is bounded by

$$C_h \exp\left(\sum_{j \in \mathcal{I}} (\lambda_j + \varepsilon \operatorname{sign} z_j) z_j \right). \tag{3.2}$$

This is integrable in each negative direction. For positive directions, put $F = \{j \in \mathcal{I} : z_j > 1\}$ and $Z_F = \sum_{j \in F} z_j$. Select a descendant $i_j \in \mathcal{B}_j$ in each active group. Its path $Y_{ji_j,L}$ is a standard Brownian motion on $[0, h]$, and selected paths are independent across groups. A crossing implies

$$\sup_{0 \le s \le h} \sum_{j \in F} Y_{ji_j,L}(s) > Z_F - C_h.$$

The reflection principle therefore bounds the conditional crossing probability by $C_h \exp(-((Z_F - C_h)_+)^2/(2mh))$ when $F \neq \varnothing$. On each of the finitely many regions determined by F, this Gaussian decay and (3.2) provide an integrable envelope. Dominated convergence and $v_j = \exp(\lambda_j z_j)$ now yield

$$\frac{M_u(L)}{p_T(u)} \longrightarrow \exp(-hQ_*/2) \mathbb{E}\left\{ \operatorname{Leb}_m \left(\bigcup_{0 \le s \le h} \prod_{j \in \mathcal{I}} (0, \exp\left(\lambda_j \min_{i \in \mathcal{B}_j} (Y_{ji,L}(s)) \right)) \right) \right\}. \tag{3.3}$$

The Jacobian is $\exp(\boldsymbol{\lambda}_{\mathcal{I}}^\top \boldsymbol{z})\, \mathrm{d}\boldsymbol{z} = (\prod_{j \in \mathcal{I}} \lambda_j)^{-1}\, \mathrm{d}\boldsymbol{v}$, which cancels the active-tail prefactor.
Subtract $U_j(L)$ from each path in group j. The volume in (3.3) acquires the multiplicative factor $\exp\left(\sum_{j \in \mathcal{I}} \lambda_j U_j(L)\right)$. Tilt by the mean-one density $\exp\left(\sum_{j \in \mathcal{I}} \lambda_j U_j(L) - LQ_*/2\right)$. Under this tilt, $(U_j(L - s) - U_j(L))_{0 \le s \le L}$ has the law $(W_j(s) - \lambda_j s)_{0 \le s \le L}$. The genealogies are

unchanged and independent of the reversed paths. Hence the limit in (3.3) is precisely

$$C_{\mathcal{B},L} = \exp(-\kappa Q_*/2)\mathbb{E}\left\{ \mathrm{Leb}_m \left(\bigcup_{-L\le t\le \kappa} \prod_{j\in\mathcal{I}} (0, \exp\left(\lambda_j \min_{i\in\mathcal{B}_j}(\xi_{ji}(t))\right)) \right) \right\}. \tag{3.4}$$

We next justify localisation. For the skeleton and any $0 < S \le T$, a stopping-time comparison yields

(3.5)

$$\mathbb{P}\{\exists t \le S : \boldsymbol{B}_{\mathcal{I}}(t) > u\boldsymbol{q}_{\mathcal{I}} + \boldsymbol{a}_{\mathcal{I}}(t)\} \le K_{\mathcal{I}}\mathbb{P}\{\boldsymbol{B}_{\mathcal{I}}(S) > u\boldsymbol{q}_{\mathcal{I}} + \boldsymbol{a}_{\mathcal{I}}(S)\}, \quad K_{\mathcal{I}} = \prod_{j\in\mathcal{I}} \max(2, \overline{\Phi}(c_j\sqrt{T})^{-1}).$$

Here $\overline{\Phi}$ is the standard normal upper tail. To see this, stop when the minimum of the active margins first reaches $\epsilon > 0$. Conditional future increments have nonnegative Gaussian covariances. Gaussian association bounds their joint probability of exceeding the linear trend increments below by the product of the marginal probabilities, each at least $\min(1/2, \overline{\Phi}(c_j\sqrt{T}))$. The endpoint at the stopping time is already above its barrier. Let $\epsilon \downarrow 0$ to obtain (3.5); this is Lemma A.7. Since $S_u < r_u$, a grouped crossing before S_u implies the skeleton active event in (3.5). A Gaussian tail calculation using A_u yields

$$\frac{\mathbb{P}\{\boldsymbol{B}_{\mathcal{I}}(S_u) > u\boldsymbol{q}_{\mathcal{I}} + \boldsymbol{a}_{\mathcal{I}}(S_u)\}}{p_T(u)} \longrightarrow \pi_{\mathcal{J}}(\boldsymbol{b}_{\mathcal{J}})^{-1} \exp(-hQ_*/2).$$

Consequently we have

$$\limsup_{u\to\infty} \frac{\Psi_{\mathcal{B}}(u) - M_u(L)}{p_T(u)} \le K_{\mathcal{I}}\pi_{\mathcal{J}}(\boldsymbol{b}_{\mathcal{J}})^{-1} \exp(-(L+\kappa)Q_*/2). \tag{3.6}$$

By monotone convergence, $C_{\mathcal{B},L} \uparrow C_{\mathcal{B}}$. The constant is finite: the past volume is bounded by a product of exponentials of individual drifted Brownian suprema, whose expectation is 2^m. The future volume is bounded by a product of exponentials of Brownian suprema on $[0,\kappa]$, which has finite expectation. The box at $t = 0$ has volume one and hence $C_{\mathcal{B}} > 0$. Letting $L \to \infty$ in (3.6) proves the ruin assertion.

For the terminal event, condition on $\boldsymbol{B}_{\mathcal{I}}(r_u) = u\boldsymbol{q}_{\mathcal{I}} + \boldsymbol{a}_{\mathcal{I}} - \boldsymbol{z}/u$, using the terminal shift $\boldsymbol{a}_{\mathcal{I}} = \boldsymbol{a}_{\mathcal{I}}(T)$. The active density ratio converges to $\exp(\boldsymbol{\lambda}_{\mathcal{I}}^\top \boldsymbol{z} - \kappa Q_*/2)$, the inactive factor is again $\pi_{\mathcal{J}}(\boldsymbol{b}_{\mathcal{J}})$, and the active event is $L_j > z_j$ for all $j \in \mathcal{I}$. The same domination argument applies, using one Gaussian endpoint per group for positive directions. Tonelli's theorem yields

$$\left(\prod_{j\in\mathcal{I}} \lambda_j\right) \int_{\mathbb{R}^m} \exp(\boldsymbol{\lambda}_{\mathcal{I}}^\top \boldsymbol{z})\mathbb{P}\{L_j > z_j,\ j \in \mathcal{I}\}\,\mathrm{d}\boldsymbol{z} = \mathbb{E}\left\{\exp\left(\sum_{j\in\mathcal{I}} \lambda_j L_j\right)\right\}.$$

Independence across the genealogies yields (2.6). Its positivity and finiteness follow immediately. This proves the remaining assertions. □

Proof of Corollary 2.2. For singleton groups, write G_j and ξ_j for the unique terminal trajectory and local field in group j. For sufficiently large u, the cut r_u lies in every terminal skeleton edge. The increments after the cut are independent Brownian motions, independent of the past. For singleton groups, the graft replaces these increments by Brownian motions with exactly the same law, since $u^{-1}G_j(u^2 t)$ is standard Brownian motion. This proves equality in law of the entire processes, including their common crossing time.

The terminal identity is also immediate from the factorisation:

$$D_{\mathcal{B}} = \exp(-\kappa Q_*/2) \prod_{j\in\mathcal{I}} \mathbb{E}\{\exp(\lambda_j G_j(\kappa))\} = \exp(-\kappa Q_*/2) \prod_{j\in\mathcal{I}} \exp(\kappa\lambda_j^2/2) = 1.$$

We verify the ruin identity directly at the level of its constant. Define

$$\mathcal{V}(f) = \mathrm{Leb}_m\left(\bigcup_{t\le\kappa}\prod_{j\in\mathcal{I}}(0,\exp(\lambda_j f_j(t)))\right).$$

Translating each path by its terminal value yields

$$\mathcal{V}(\xi) = \exp\left(\sum_{j\in\mathcal{I}}\lambda_j G_j(\kappa)\right)\mathcal{V}(\xi - G(\kappa)).$$

Under the probability measure with density

$$\frac{d\widehat{\mathbb{P}}}{d\mathbb{P}} = \exp\left(\sum_{j\in\mathcal{I}}\lambda_j G_j(\kappa) - \kappa Q_*/2\right),$$

each G_j has drift λ_j, while the original reversed trunks W_j remain independent standard Brownian motions. Set $\widehat{\xi}_j(s) = \xi_j(\kappa - s) - G_j(\kappa)$, $s \ge 0$. On $[0,\kappa]$ this is the reversed terminal segment, a Brownian motion with drift $-\lambda_j$. For $s \ge \kappa$ it continues as

$$\widehat{\xi}_j(s) = -G_j(\kappa) + W_j(s-\kappa) - \lambda_j(s-\kappa),$$

with independent Brownian increments and the same drift. Hence under $\widehat{\mathbb{P}}$, the processes $\widehat{\xi}_j$, $j \in \mathcal{I}$, are independent Brownian motions with drifts $-\lambda_j$ on $[0,\infty)$. It follows that

$$C_{\mathcal{B}} = \mathbb{E}_{\widehat{\mathbb{P}}}\left\{\mathrm{Leb}_m\left(\bigcup_{s\ge 0}\prod_{j\in\mathcal{I}}(0,\exp(\lambda_j\widehat{\xi}_j(s)))\right)\right\} = \mathcal{H}_{\mathcal{I}}(\boldsymbol{\lambda}_{\mathcal{I}}).$$

The equality with $\mathcal{H}_{\mathcal{I}}(\boldsymbol{\lambda}_{\mathcal{I}})$ follows from Tonelli's theorem and the change of variables $y_j = \exp(z_j)$ in its defining integral. Substitution in Theorem 2.1 proves the corollary. □

Proposition 3.1. *The constants satisfy*

$$\exp(-\kappa Q_*/2)\mathcal{H}_{\mathcal{I}}(\boldsymbol{\lambda}_{\mathcal{I}}) \le C_{\mathcal{B}} \le \mathcal{H}_{\mathcal{I}}(\boldsymbol{\lambda}_{\mathcal{I}}), \qquad 0 < D_{\mathcal{B}} \le 1, \qquad D_{\mathcal{B}} \le C_{\mathcal{B}}. \tag{3.7}$$

For the truncation in (3.4)

$$0 \le C_{\mathcal{B}} - C_{\mathcal{B},L} \le \exp(-(\kappa+L)Q_*/2)\mathcal{H}_{\mathcal{I}}(\boldsymbol{\lambda}_{\mathcal{I}}) \le 2^m\exp(-(\kappa+L)Q_*/2). \tag{3.8}$$

Proof. The past alone yields the lower bound for $C_{\mathcal{B}}$; the box at $t = \kappa$ yields $C_{\mathcal{B}} \ge D_{\mathcal{B}}$. Selecting one representative leaf per group yields the upper bound. By the singleton identity (2.10), the representative constant is $\mathcal{H}_{\mathcal{I}}(\boldsymbol{\lambda}_{\mathcal{I}})$. Also $L_j \le G_{ji}(\kappa)$ for a selected leaf, so $\mathbb{E}\{\exp(\lambda_j L_j)\} \le \exp(\kappa\lambda_j^2/2)$ and $D_{\mathcal{B}} \le 1$.
The omitted boxes lie on the past segment $s \ge L$. Factor their coordinate widths at time L. Independent future increments yield

$$\mathbb{E}\left\{\mathrm{Leb}_m\left(\bigcup_{s\ge L}\prod_{j\in\mathcal{I}}(0,\exp(\lambda_j W_j(s) - \lambda_j^2 s))\right)\right\} = \exp(-LQ_*/2)\mathcal{H}_{\mathcal{I}}(\boldsymbol{\lambda}_{\mathcal{I}}).$$

Multiplication by $\exp(-\kappa Q_*/2)$ proves (3.8). □

The error in finite-time simulation caused by omitting the remote part of the reversed trunk is therefore controlled. On $[-L,\kappa]$, nested time grids with dense union recover the exact union-of-boxes volume monotonically, by path continuity and strict inequalities. Finite-window volumes have moments of every order. Their expectations can therefore be evaluated by simulation of the finitely many Brownian edge paths and geometric union volumes. The bound (3.8) controls time truncation only; grid discretisation and sampling errors require separate control. The untruncated functional need not have finite variance, as the one-group case without a genealogy shows.

The singleton identities (2.10) also hold under the degenerate convention $\kappa = 0$, with every group collapsed to its trunk.

Example 3.2. *Fix $\kappa > 0$ and a finite rooted metric tree $\mathcal{G}$ with positive edge lengths, root height zero, and $d_{\mathcal{G}}$ leaves at height κ. Let $G_i(s)$, $1 \le i \le d_{\mathcal{G}}$, be its leaf trajectories, obtained from independent standard Brownian motions on its edges. Put*

$$M_{\mathcal{G}} = \sup_{0\le s\le\kappa} \min_{1\le i\le d_{\mathcal{G}}} (G_i(s)) \ge 0, \qquad L_{\mathcal{G}} = \min_{1\le i\le d_{\mathcal{G}}} (G_i(\kappa)).$$

If this is the only active terminal group and its multiplier is $\lambda > 0$, write $C_{\mathcal{G}}, D_{\mathcal{G}}$ for its constants. Let E_λ, independent of the genealogy, satisfy $\mathbb{P}\{E_\lambda > y\} = \exp(-2\lambda y)$, $y \ge 0$. Then

$$C_{\mathcal{G}} = \exp(-\kappa\lambda^2/2)\mathbb{E}\{\exp(\lambda \max(E_\lambda, M_{\mathcal{G}}))\} = 2\exp(-\kappa\lambda^2/2)\mathbb{E}\{\cosh(\lambda M_{\mathcal{G}})\},$$
$$D_{\mathcal{G}} = \exp(-\kappa\lambda^2/2)\mathbb{E}\{\exp(\lambda L_{\mathcal{G}})\}.$$

Indeed, for $m \ge 0$, $\mathbb{E}\{\exp(\lambda \max(E_\lambda, m))\} = \exp(\lambda m) + \exp(-\lambda m)$. Conditioning at the split, E_λ represents the limiting backward maximum of the rescaled trunk and $M_{\mathcal{G}}$ the forward genealogy.

For the single-branch skeleton with $T = 1$, zero drifts and shifts, and common barrier u, we have $\lambda = 1$. Let W_0 be its Brownian trunk, independent of $\mathcal{G}$, and put $r_u = 1 - \kappa/u^2$ for $u > \sqrt{\kappa}$. The descendant paths are $B_i^{(u)} = X_{1i}^{(u)}$ from (2.1), with $B_1 = W_0$ and $G_{1i} = G_i$. Denote their simultaneous ruin and terminal probabilities by $\Psi_{\mathcal{G}}(u)$ and $p_{\mathcal{G}}(u)$ as in (2.2)–(2.3). Theorem 2.1 yields

$$\begin{aligned} \Psi_{\mathcal{G}}(u) &\sim 2\exp(-\kappa/2)\mathbb{E}\{\cosh(M_{\mathcal{G}})\}\overline{\Phi}(u), \\ p_{\mathcal{G}}(u) &\sim \exp(-\kappa/2)\mathbb{E}\{\exp(L_{\mathcal{G}})\}\overline{\Phi}(u), \end{aligned} \tag{3.9}$$

and therefore

$$\frac{\Psi_{\mathcal{G}}(u)}{p_{\mathcal{G}}(u)} \longrightarrow \frac{2\mathbb{E}\{\cosh(M_{\mathcal{G}})\}}{\mathbb{E}\{\exp(L_{\mathcal{G}})\}}.$$

The height, scaled branching times, and number of leaves are fixed. For a single terminal branch the two coefficients are 2 and 1. For two independent terminal edges, $M_{\mathcal{G}} \overset{d}{=} M_\kappa$, and with $E = E_1$ the coefficient becomes $C_\kappa = \exp(-\kappa/2)\mathbb{E}\{\exp(\max(E, M_\kappa))\}$. This recovers the path asymptotic in Lemma 3.12. Section A contains the independent reflection proof of (3.9) and the detailed Brownian-bridge conditioning argument.

For several active groups, separate exponential trunk maxima do not preserve their crossing times and cannot replace the common-time backward curve in (2.5). The terminal coefficient factors, whereas the ruin coefficient generally does not. Theorem 2.1 permits different fixed barriers and drifts between groups; changing scaled genealogies, growing leaf counts, and descendant-specific local barriers require further estimates.

Example 3.3. *Let $T = 1$, take zero trends and shifts, and take the barrier u for every leaf. The skeleton has two leaves whose common trunk ends at time $\rho \in [0, 1)$, so*

$$\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \qquad \boldsymbol{q} = (1,1)^\top, \qquad \mathcal{I} = \{1,2\}, \qquad \lambda_1 = \lambda_2 = \lambda = \frac{1}{1+\rho}.$$

For fixed $\kappa > 0$ and all sufficiently large u, at $r_u = 1 - \kappa/u^2$ each skeleton branch splits into two independent terminal edges. The four final leaves form the sister pairs $\mathcal{B}_1 = \{1,2\}$ and $\mathcal{B}_2 = \{3,4\}$. Each pair shares its value at r_u, so conditioning produces two level offsets, one per pair. Figure 2 illustrates the construction. The four edge processes $G_{11}, G_{12}, G_{23}, G_{24}$ on $[0,\kappa]$ are independent standard Brownian motions. Here $\pi_{\mathcal{J}}(\boldsymbol{b}_{\mathcal{J}}) = 1$, $Q_* = 2\lambda^2$, and the density correction is $\exp(-\kappa\lambda^2)$.

For independent standard normal Z_1, Z_2

$$\mathbb{E}\{\exp(a\min(Z_1, Z_2))\} = 2\exp(a^2/2)\overline{\Phi}(a/\sqrt{2}), \qquad a \ge 0.$$

Each sister pair contributes one such terminal factor. Therefore

$$D_{\mathcal{B}} = \exp(-\kappa\lambda^2)\left(2\exp(\kappa\lambda^2/2)\overline{\Phi}\big(\lambda\sqrt{\kappa/2}\big)\right)^2 = 4\overline{\Phi}\big(\lambda\sqrt{\kappa/2}\big)^2. \tag{3.10}$$

To specify the ruin coefficient, put

$$F_j(t) = \begin{cases} W_j(-t) - \lambda(-t), & t \le 0, \\ \min_{i\in\mathcal{B}_j}(G_{ji}(t)), & 0 \le t \le \kappa, \end{cases} \qquad j = 1, 2,$$

where W_1, W_2 are independent Brownian motions, independent of the terminal edges. Then

$$C_{\mathcal{B}} = \exp(-\kappa\lambda^2)\mathbb{E}\left\{\mathrm{Leb}_2\left(\bigcup_{t\le\kappa}((0,\exp(\lambda F_1(t))) \times (0,\exp(\lambda F_2(t))))\right)\right\}. \tag{3.11}$$

The rectangles use the same time in both coordinates, since both pairs must cross simultaneously. Thus the terminal coefficient factors over the pairs, whereas the ruin coefficient uses their common crossing time.
The skeleton reference tail is

$$p_1(u) \sim \frac{(1+\rho)^2}{2\pi\sqrt{1-\rho^2}} u^{-2}\exp(-u^2/(1+\rho)).$$

Theorem 2.1 yields $p_{\mathcal{B}}(u) \sim D_{\mathcal{B}}p_1(u)$ and $\Psi_{\mathcal{B}}(u) \sim C_{\mathcal{B}}p_1(u)$, hence the limiting ruin-to-terminal ratio is $C_{\mathcal{B}}/D_{\mathcal{B}}$. In particular, the power u^{-2} is determined by the two groups, although the event involves four leaves. At $\rho = 1/2$ and $\kappa = 1$

$$\lambda = \frac{2}{3}, \qquad \exp(-\kappa\lambda^2) = \exp(-4/9), \qquad D_{\mathcal{B}} = 4\overline{\Phi}(\sqrt{2}/3)^2 \approx 0.406217.$$

The ruin coefficient can be approximated by the rectangle-union procedure described above. Brownian time scaling and (3.8) yield

$$\exp(-\kappa\lambda^2)\mathcal{H}_2 \le C_{\mathcal{B}} \le \mathcal{H}_2, \qquad 0 \le C_{\mathcal{B}} - C_{\mathcal{B},L} \le \mathcal{H}_2\exp(-(\kappa+L)\lambda^2) \le 4\exp(-(\kappa+L)\lambda^2).$$

Replacing either pair by a single leaf yields the three-leaf model with $D_{\mathcal{B}} = 2\overline{\Phi}(\lambda\sqrt{\kappa/2})$. If both groups are singletons, $D_{\mathcal{B}} = 1$ and $C_{\mathcal{B}} = \mathcal{H}_2$, recovering the fixed-tree case. More generally, an immediate star with d_j descendants yields the one-dimensional integral

$$\mathbb{E}\{\exp(\lambda_j L_j)\} = d_j\int_{\mathbb{R}}\exp(\lambda_j\sqrt{\kappa}z)\phi(z)\overline{\Phi}(z)^{d_j-1}\,\mathrm{d}z,$$

so terminal coefficients for several stars remain directly computable.

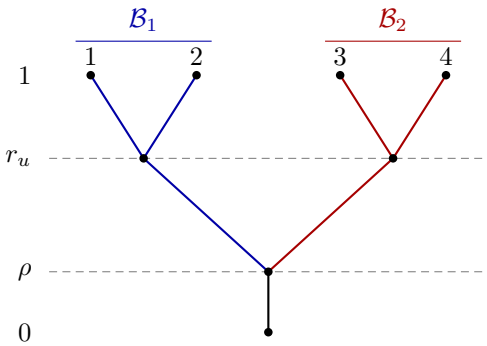


FIGURE 2. Two terminal sister pairs. The first split at ρ is fixed; the two late splits occur at $r_u = 1 - \kappa/u^2$. Heights are schematic.

The active set can be computed from voltages and currents on the tree. We regard an edge e as a resistor of resistance ℓ_e, and use the Dirichlet energy

$$\mathcal{D}(f) = \sum_{e=(v,w)\in\mathcal{E}} \frac{(f(w)-f(v))^2}{\ell_e}, \qquad f : \mathcal{V} \to \mathbb{R}, \quad f(\rho) = 0.$$

Currents on edges are oriented towards the root.

Proposition 3.4. *The quadratic cost satisfies*

$$Q_\Sigma(\boldsymbol{q}) = \min_{\substack{f:\mathcal{V}\to\mathbb{R} \\ f(\rho)=0,\ f(i)\geq q_i,\ i\in\mathcal{L}}} (\mathcal{D}(f)).$$

The minimiser $f_\star$ is unique, is harmonic at every non-root, non-leaf vertex, and has leaf values $(f_\star(i))_{i\in\mathcal{L}} = \widetilde{\boldsymbol{q}}$. Its terminal currents satisfy

$$\kappa_i(f_\star) := \frac{f_\star(i) - f_\star(p(i))}{\ell_{(p(i),i)}} = \lambda_i \geq 0,$$
$$\lambda_i(f_\star(i) - q_i) = 0.$$

Thus the active leaves are precisely those carrying positive current:

$$\mathcal{I} = \{i \in \mathcal{L} : \kappa_i(f_\star) > 0\}, \qquad Q_\Sigma(\boldsymbol{q}) = \sum_{i\in\mathcal{I}} q_i\lambda_i.$$

If $\mathcal{L}(w)$ is the set of leaves descending from w, then

$$\frac{f_\star(w) - f_\star(v)}{\ell_{(v,w)}} = \sum_{i\in\mathcal{L}(w)} \lambda_i, \qquad (v,w) \in \mathcal{E}.$$

In particular, $f_\star$ is non-decreasing along each root-to-leaf path, and $f_\star(i) = f_\star(p(i))$ for every $i \in \mathcal{J}$.

Proof of Proposition 3.4. Let P be the leaf-edge incidence matrix, where $P_{ie} = 1$ if edge e lies on the path from ρ to i, and $P_{ie} = 0$ otherwise. Put $\Lambda = \mathrm{diag}(\ell_e : e \in \mathcal{E})$. The edge-increment construction yields $\Sigma = P\Lambda P^\top$. For a voltage f, write $z_e = f(w) - f(v)$ on $e = (v,w)$. Since the graph is a tree, these increments determine f uniquely when $f(\rho) = 0$; prescribed leaf values $\boldsymbol{y}$ are equivalent to $P\boldsymbol{z} = \boldsymbol{y}$. Quadratic minimisation therefore yields

$$\min_{P\boldsymbol{z}=\boldsymbol{y}} (\boldsymbol{z}^\top\Lambda^{-1}\boldsymbol{z}) = \boldsymbol{y}^\top\Sigma^{-1}\boldsymbol{y}, \qquad \boldsymbol{z}_\star = \Lambda P^\top\Sigma^{-1}\boldsymbol{y}.$$

Equivalently

$$\min_{\substack{f(\rho)=0 \\ f(i)=y_i,\ i\in\mathcal{L}}} (\mathcal{D}(f)) = \boldsymbol{y}^\top\Sigma^{-1}\boldsymbol{y}. \tag{3.12}$$

Minimising over $\boldsymbol{y} \geq \boldsymbol{q}$ proves the obstacle identity and identifies its leaf values as $\widetilde{\boldsymbol{q}}$. The energy is strictly convex on $\{f : f(\rho) = 0\}$: a zero-energy function is constant on the connected tree and hence zero. This also proves uniqueness.

Variation at an interior vertex yields harmonicity. The minimising edge increment formula shows that the current through (v,w), towards the root, equals $\sum_{i\in\mathcal{L}(w)}(\Sigma^{-1}\boldsymbol{y})_i$. At $\boldsymbol{y} = \widetilde{\boldsymbol{q}}$, Lemma A.1 identifies these leaf currents as $\boldsymbol{\lambda}$ and establishes their non-negativity and complementarity. Finally, $Q_\Sigma(\boldsymbol{q}) = \widetilde{\boldsymbol{q}}^\top\boldsymbol{\lambda} = \boldsymbol{q}^\top\boldsymbol{\lambda}$. □

The active set can be found by the following leaf-release procedure.

Lemma 3.5. *For a non-empty set $A \subset \mathcal{L}$, let f_A minimise $\mathcal{D}(f)$ subject to*

$$f(\rho) = 0, \qquad f(i) = q_i \quad (i \in A),$$

with all other vertices free. Put

$$\kappa_i(A) = \frac{q_i - f_A(p(i))}{\ell_{(p(i),i)}}, \qquad i \in A.$$

Start with $A = \mathcal{L}$. If some $j \in A$ has $\kappa_j(A) \le 0$, release that leaf by replacing A with $A \setminus \{j\}$, and recompute the harmonic voltage. Stop when every remaining pinned leaf has positive current.
The procedure makes at most $d-1$ releases, and at termination

$$A = \mathcal{I}, \qquad f_A = f_\star, \qquad \kappa_i(A) = \lambda_i \quad (i \in A).$$

The result is independent of the release order. Zero-current leaves are released even when their final voltage equals their obstacle. Harmonic elimination on the tree requires $O(d|\mathcal{V}|)$ arithmetic operations and $O(|\mathcal{V}|)$ working storage.

Proof of Lemma 3.5. For adjacent vertices write ℓ_{vw} for their edge length and set

$$Lf(v) = \sum_{w : w \sim v} \frac{f(v) - f(w)}{\ell_{vw}}.$$

Strict convexity ensures a unique minimiser for each pinned set, and

$$Lf_A(v) = 0, \qquad v \in \mathcal{V} \setminus (\{\rho\} \cup A).$$

In particular, every released leaf has zero terminal current.
Suppose $j \in A$, $\kappa_j(A) \le 0$, and put $A' = A \setminus \{j\}$ and $h = f_{A'} - f_A$. The root condition ensures that $f_{A'}$ exists even if A' is empty. The function h vanishes on $\{\rho\} \cup A'$, is harmonic at previously free vertices, and satisfies

$$Lh(j) = -\kappa_j(A) \ge 0.$$

The discrete minimum principle implies $h \ge 0$. Indeed, a negative minimum at a free vertex would force all its neighbours to share that minimum, and connectedness would propagate it to a fixed vertex, where $h = 0$. Thus releasing a nonpositive-current leaf cannot decrease any voltage. A zero-current release leaves the voltage unchanged, by uniqueness.
Inductively, all released leaves satisfy their obstacles. At the moment of release the new leaf has voltage at least its previous value q_j, and previously released voltages cannot decrease. Moreover, discrete summation by parts yields, for every non-empty A

$$\mathcal{D}(f_A) = \sum_{i \in A} q_i \kappa_i(A) > 0.$$

The energy is positive because a positive leaf value is pinned while the root has value zero. Since every $q_i > 0$, at least one pinned current is positive. A release therefore leaves a non-empty pinned set, and there can be at most $d-1$ releases.
At termination, all leaf obstacles are satisfied. The pinned leaves attain their obstacles and have positive current, the released leaves have zero current, and all interior vertices are harmonic. These are the optimality conditions in Proposition 3.4, so $f_A = f_\star$ and $A = \mathcal{I}$. Uniqueness proves independence of the release order. Releasing zero-current leaves identifies the positive-multiplier set; it does not change the minimising voltage at that step.
For the computational bound, fixing the root and pinned leaves leaves a positive definite linear system whose graph is a forest. Successively eliminating vertices of degree at most one in each free component creates no additional edges and has positive pivots. Elimination and back-substitution take $O(|\mathcal{V}|)$ operations and storage. Computing the terminal currents has the same bound. There are at most d harmonic solves including the initial solve. □

The network interpretation of $\mathbf{1}_d^\top \Sigma^{-1} \mathbf{1}_d$ also identifies the active set in the common-barrier case.

Remark 3.6. *Regard each edge e as a resistor of resistance ℓ_e. Hold the root at potential 0 and all leaves at potential 1. If $Q(\mathcal{T})$ is the effective conductance between the root and the terminal boundary, then*

$$Q(\mathcal{T}) = \mathbf{1}_d^\top \Sigma^{-1} \mathbf{1}_d, \qquad \Sigma^{-1}\mathbf{1}_d > \mathbf{0}.$$

Moreover, $(\Sigma^{-1}\mathbf{1}_d)_i$ is the current entering through leaf i.
Pinning all leaves at potential 1, the harmonic voltage satisfies $0 \le f(p(i)) < 1$ by the maximum principle. Thus every terminal current $(1 - f(p(i)))/\ell_{(p(i),i)}$ is positive. The energy identity in Proposition 3.4 now yields the result.

For $\boldsymbol{q} = \mathbf{1}_d$, Remark 3.6 yields $\boldsymbol{\lambda} = \Sigma^{-1}\mathbf{1}_d > \mathbf{0}$. Hence the active set is full. Consequently, for $\boldsymbol{q} = \mathbf{1}_d$ and $c_1 = \cdots = c_d = c$

$$\begin{aligned}&\mathbb{P}\{\exists t \in [0,T] : B_i(t) - ct > u,\ i = 1, \dots, d\}\\ &\sim \mathcal{H}_{\{1,\dots,d\}}(\boldsymbol{\lambda}) \frac{u^{-d}}{(2\pi)^{d/2}|\Sigma|^{1/2}\prod_{i=1}^d \lambda_i} \exp\left(-\frac{1}{2}(u+cT)^2 Q(\mathcal{T})\right),\end{aligned}$$

where

$$\boldsymbol{\lambda} = \Sigma^{-1}\mathbf{1}_d, \qquad Q(\mathcal{T}) = \mathbf{1}_d^\top \Sigma^{-1}\mathbf{1}_d.$$

The conductance can be computed without forming Σ. If C_v is the conductance from a vertex v to its descendant leaves held at a common potential, the series and parallel rules yield

$$\begin{aligned}C_i &= \infty \quad (i \in \mathcal{L}), \qquad Q(\mathcal{T}) = C_\rho,\\ C_v &= \sum_{w:(v,w)\in\mathcal{E}} \left(\ell_{(v,w)} + C_w^{-1}\right)^{-1} \quad (v \notin \mathcal{L}),\end{aligned}$$

with $\infty^{-1} = 0$.

Example 3.7. *Let $T = 3$. After a common trunk of length 1, leaf 1 follows a terminal edge of length 2. The other branch continues to height 2 and splits into leaves 2 and 3, each with terminal length 1. For $\boldsymbol{q} = \mathbf{1}_3$*

$$\Sigma = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 3 & 2 \\ 1 & 2 & 3 \end{pmatrix}, \qquad \boldsymbol{\lambda} = \frac{1}{13}\begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix}.$$

Thus $Q(\mathcal{T}) = 7/13$ and $|\Sigma| = 13$. The terminal currents are positive but unequal. The two sibling leaves carry current $2/13$ each; the remaining leaf carries current $3/13$. For common drift c and zero shifts

$$\Psi_{\mathcal{T}}(u) \sim \mathcal{H}_{\{1,2,3\}}(\boldsymbol{\lambda}) \frac{13^{5/2}}{12(2\pi)^{3/2}} u^{-3} \exp\left(-\frac{7}{26}(u+3c)^2\right).$$

The weights entering the local constant are unequal despite the common barrier.

Example 3.8. *Fix $n \ge 1$. Let each particle split into two at times $1, \dots, n$, and let $T_n = n+1$ and $d_n = 2^n$. For $\boldsymbol{q} = \mathbf{1}_{d_n}$, symmetry yields*

$$\Sigma_n^{-1}\mathbf{1}_{d_n} = \lambda_n \mathbf{1}_{d_n}, \qquad \lambda_n = (2^{n+1}-1)^{-1}, \qquad Q_n = \frac{2^n}{2^{n+1}-1}.$$

The remaining eigenvalues and their multiplicities are

$$\mu_{v,n} = 2^{n-v+1} - 1, \qquad 2^{v-1}, \qquad v = 1, \dots, n,$$

so

$$|\Sigma_n| = (2^{n+1}-1)\prod_{v=1}^n \mu_{v,n}^{2^{v-1}}.$$

These identities follow from the constant vector and the generation-wise Haar contrasts, as in [2, Proposition 2.4]. Scale invariance yields $\mathcal{H}_{\{1,\dots,d_n\}}(\lambda_n \mathbf{1}_{d_n}) = \mathcal{H}_{\{1,\dots,d_n\}}(\mathbf{1}_{d_n})$. Substitution into Corollary 2.2 recovers [2, Example 3.8]. The depth n remains fixed in this specialisation.

Example 3.9. *Let two leaves separate at time $r \in (0,T)$, put $\varrho = r/T$, and take $\boldsymbol{q} = (1,a)^\top$, where $0 < a \le 1$, with $\boldsymbol{c} = \boldsymbol{x} = \boldsymbol{0}$. Then*

$$\Sigma = T \begin{pmatrix} 1 & \varrho \\ \varrho & 1 \end{pmatrix}.$$

Write $\Psi_a(u)$ for the simultaneous exceedance probability and $\overline{\Phi}$ for the standard normal survival function. If $a < \varrho$, then

$$\mathcal{I} = \{1\}, \qquad \widetilde{\boldsymbol{q}} = (1,\varrho)^\top, \qquad \lambda_1 = Q_\Sigma(\boldsymbol{q}) = T^{-1}, \qquad \pi_{\mathcal{J}} = 1,$$

and hence

$$\Psi_a(u) \sim 2\overline{\Phi}(u/\sqrt{T}).$$

At $a = \varrho$, the active set is still $\{1\}$, but the second constraint is binding with zero multiplier. Since

$$B_2(T) \mid B_1(T) = y \sim N\big(\varrho y, T(1-\varrho^2)\big),$$

its inactive boundary factor is $\pi_{\mathcal{J}} = 1/2$. Therefore

$$\Psi_\varrho(u) \sim \overline{\Phi}(u/\sqrt{T}).$$

If $a > \varrho$, both coordinates are active and

$$\boldsymbol{\lambda} = \frac{1}{T(1-\varrho^2)} \begin{pmatrix} 1-\varrho a \\ a-\varrho \end{pmatrix}, \qquad Q_\Sigma(\boldsymbol{q}) = \frac{1}{T} + \frac{(a-\varrho)^2}{T(1-\varrho^2)}.$$

Consequently we have

$$\Psi_a(u) \sim \mathcal{H}_{\{1,2\}}(\boldsymbol{\lambda}) \frac{T(1-\varrho^2)^{3/2}}{2\pi(1-\varrho a)(a-\varrho)} u^{-2} \exp(-Q_\Sigma(\boldsymbol{q})u^2/2).$$

These assertions follow from Lemma A.1 and Corollary 2.2, using $\mathcal{H}_{\{1\}}(T^{-1}) = 2$. In the leaf-release procedure, the initial second current has the sign of $a - \varrho$. Releasing leaf 2 when $a \le \varrho$ leaves its potential at ϱ; equality yields the zero-current binding case. For the concrete choice $T = 1$ and $r = 1/2$, the first two regimes read

$$\mathbb{P}\{\exists t \in [0,1] : B_1(t) > u,\ B_2(t) > au\} \sim \begin{cases} 2\overline{\Phi}(u), & 0 < a < 1/2, \\ \overline{\Phi}(u), & a = 1/2. \end{cases}$$

Conditioning on $B_1(1)$ near u centres $B_2(1)$ near $u/2$. The lower barrier is asymptotically redundant, while the barrier at the conditional mean contributes the factor $1/2$.

Example 3.10. *Take $T = 1$, let the two leaves separate at time $1/2$, and fix $c \in \mathbb{R}$. Then*

$$\begin{aligned} &\mathbb{P}\{\exists t \in [0,1] : B_1(t) > u,\ B_2(t) - ct > u/2\} \\ &\sim 2\overline{\Phi}\left(\frac{2c}{\sqrt{3}}\right)\overline{\Phi}(u). \end{aligned} \tag{3.13}$$

Indeed, the terminal parameters are

$$\Sigma = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}, \qquad \boldsymbol{q} = \begin{pmatrix} 1 \\ 1/2 \end{pmatrix}, \qquad \boldsymbol{a} = \begin{pmatrix} 0 \\ c \end{pmatrix}.$$

The active set is $\mathcal{I} = \{1\}$, with $\lambda_1 = 1$, while $b_2 = c$ and $\mathrm{Var}(B_2(1) \mid B_1(1)) = 3/4$. Consequently we have

$$\pi_{\{2\}}(c) = \overline{\Phi}(2c/\sqrt{3}), \qquad p_1(u) \sim \pi_{\{2\}}(c)\overline{\Phi}(u).$$

Corollary 2.2 and $\mathcal{H}_{\{1\}}(1) = 2$ yield (3.13). At $c = 0$, this recovers the boundary case in Example 3.9; a positive c decreases its leading constant. The trend disappears from the local Brownian fluctuation limit but remains in the inactive Gaussian boundary factor. The formula contains only ordinary normal tails.

Remark 3.11. *For the two-leaf tree in Example 3.9 and fixed $\theta \in \mathbb{R}$, set*

$$\Psi_\theta^{\mathrm{tr}}(u) = \mathbb{P}\{\exists t \in [0,T] : B_1(t) > u,\ B_2(t) > \varrho u + \theta\}.$$

Then

$$\frac{\Psi_\theta^{\mathrm{tr}}(u)}{2\overline{\Phi}(u/\sqrt{T})} \longrightarrow \overline{\Phi}\left(\frac{\theta}{\sqrt{T(1-\varrho^2)}}\right).$$

This follows from Corollary 2.2 with fixed $\boldsymbol{q} = (1,\varrho)^\top$, $\boldsymbol{c} = \mathbf{0}$, and $\boldsymbol{x} = (0,-\theta)^\top$, for which $b_2 = \theta$. Equivalently, the second barrier coefficient is $a_u = \varrho + \theta/u$. The assertion is for each fixed θ; it describes the transition between the fixed-barrier regimes above without asserting uniformity over unbounded offsets.

For the all-rates theorem we need more than the fixed-κ grouped asymptotic. The following two-leaf lemma establishes an exact reflection identity that will also control changing terminal lengths.

Lemma 3.12. *Fix $\kappa > 0$ and $T = 1$. For $u > \sqrt{\kappa}$, let two Brownian leaves $B_1^{(u)}, B_2^{(u)}$ share their path until*

$$r_u = 1 - \kappa/u^2$$

and have independent Brownian increments after the split. With zero drifts and common barrier u, write

$$\Psi_\kappa(u) = \mathbb{P}\Big\{\exists t \in [0,1] : B_1^{(u)}(t) > u,\ B_2^{(u)}(t) > u\Big\},$$
$$p_\kappa(u) = \mathbb{P}\Big\{B_1^{(u)}(1) > u,\ B_2^{(u)}(1) > u\Big\}.$$

With M_κ and C_κ as defined in Section 2, as $u \to \infty$ with κ fixed

$$\Psi_\kappa(u) \sim C_\kappa \overline{\Phi}(u), \qquad p_\kappa(u) \sim 2\overline{\Phi}\big(\sqrt{\kappa/2}\big)\overline{\Phi}(u). \tag{3.14}$$

In particular

$$\frac{\Psi_\kappa(u)}{p_\kappa(u)} \longrightarrow R_\kappa = \frac{C_\kappa}{D_\kappa}.$$

Moreover

$$\Psi_\kappa(u) = \mathbb{E}\left\{\overline{\Phi}\left(\frac{u - M_\kappa/u}{\sqrt{r_u}}\right) + \overline{\Phi}\left(\frac{u + M_\kappa/u}{\sqrt{r_u}}\right)\right\}, \tag{3.15}$$

and

$$M_\kappa \overset{d}{=} \sqrt{\kappa} M_1, \qquad 2\exp(-\kappa/2) \le C_\kappa \le 2.$$

Proof. Let W_0 be the common Brownian trunk and set $S_r = \sup_{0\le t\le r} W_0(t)$, with $r = r_u$. Brownian scaling on the two terminal edges yields

$$\Psi_\kappa(u) = \mathbb{P}\{S_r > u \text{ or } W_0(r) + M_\kappa/u > u\},$$

where M_κ is independent of the trunk. In particular, crossings before the split are included. Write ϕ_r for the density of $N(0,r)$. The reflection principle yields, for $x \le u$

$$\mathbb{P}\{S_r \le u,\ W_0(r) \in \mathrm{d}x\} = (\phi_r(x) - \phi_r(2u - x))\,\mathrm{d}x.$$

Conditioning on $M_\kappa = m \ge 0$ and integrating up to $u - m/u$ proves (3.15).
For passage to the limit, let $h_\kappa(z) = \mathbb{P}\{M_\kappa > z\}$, $z \ge 0$. Integrating the same killed density over endpoints below u yields

$$\Psi_\kappa(u) = 2\overline{\Phi}(u/\sqrt{r}) + \frac{1}{u}\int_0^\infty (\phi_r(u - z/u) - \phi_r(u + z/u))h_\kappa(z)\,\mathrm{d}z. \tag{3.16}$$

The first term is exactly the probability of crossing before the split. For fixed $z \geq 0$, Mills' ratio and $r = 1 - \kappa/u^2$ yield

$$\frac{\overline{\Phi}(u/\sqrt{r})}{\overline{\Phi}(u)} \to \exp(-\kappa/2), \qquad \frac{\phi_r(u \mp z/u)}{u\overline{\Phi}(u)} \to \exp(-\kappa/2 \pm z).$$

For all sufficiently large u, the nonnegative density difference in (3.16), divided by $u\overline{\Phi}(u)$, is bounded by $C\exp(2z)$. Independence and the reflection principle also yield

$$h_\kappa(z) \leq \prod_{i=1}^{2} \mathbb{P}\Big\{ \sup_{0\leq s\leq \kappa} W_i(s) > z \Big\} = 4\overline{\Phi}(z/\sqrt{\kappa})^2.$$

Thus the integrand is dominated by an integrable function. Consequently we have

$$\begin{aligned} \frac{\Psi_\kappa(u)}{\overline{\Phi}(u)} &\longrightarrow \exp(-\kappa/2)\left(2 + \int_0^\infty (\exp(z) - \exp(-z))h_\kappa(z)\,\mathrm{d}z\right) \\ &= 2\exp(-\kappa/2)\mathbb{E}\{\cosh(M_\kappa)\}. \end{aligned}$$

Since $0 \leq M_\kappa \leq \sup_{0\leq s\leq\kappa} W_1(s)$, and the latter has distribution $\sqrt{\kappa}|Z|$ for standard normal Z

$$1 \leq \mathbb{E}\{\cosh(M_\kappa)\} \leq \mathbb{E}\big\{\cosh(\sqrt{\kappa}Z)\big\} = \exp(\kappa/2).$$

This proves finiteness and the stated bounds. The scaling identity follows directly from Brownian scaling.

For the terminal tail, the average and half-difference of the two terminal values are independent centred Gaussians with variances $1 - \kappa/(2u^2)$ and $\kappa/(2u^2)$, respectively. Hence

$$p_\kappa(u) = \mathbb{E}\left\{\overline{\Phi}\left(\frac{u + \sqrt{\kappa/2}|Z|/u}{\sqrt{1-\kappa/(2u^2)}}\right)\right\}.$$

After division by $\overline{\Phi}(u)$, the integrand is at most one and converges to $\exp(-\kappa/4 - \sqrt{\kappa/2}|Z|)$. Dominated convergence and completion of the square yield

$$\frac{p_\kappa(u)}{\overline{\Phi}(u)} \longrightarrow \exp(-\kappa/4)\mathbb{E}\Big\{\exp(-\sqrt{\kappa/2}|Z|)\Big\} = 2\overline{\Phi}(\sqrt{\kappa/2}).$$

□

The split-conditioning interpretation is contained in Example 3.2; its detailed Brownian-bridge verification is given at the end of Section A.

Lemma 3.13. *The coefficients defined in Section 2 satisfy*

$$C_\kappa \longrightarrow 2, \qquad R_\kappa \longrightarrow 2, \qquad \kappa \downarrow 0,$$

and

$$C_\kappa \sim \frac{2\mathcal{H}_2}{\sqrt{\pi\kappa}}\exp(-\kappa/4), \qquad R_\kappa \longrightarrow \mathcal{H}_2, \qquad \kappa \to \infty. \tag{3.17}$$

These are limits of the coefficients obtained after taking $u \to \infty$ at fixed κ.

Proof. The small-κ limits follow from $M_\kappa \overset{d}{=} \sqrt{\kappa}M_1$, Gaussian exponential domination, and $2\overline{\Phi}(\sqrt{\kappa/2}) \to 1$. For the other limit, put $h(x) = \mathbb{P}\{M_1 > x\}$. Corollary 2.2, applied to the fixed tree consisting of two independent Brownian leaves, yields

$$h(x) \sim \mathcal{H}_2\overline{\Phi}(x)^2 \sim \frac{\mathcal{H}_2}{2\pi x^2}\exp(-x^2), \qquad x \to \infty.$$

Also $h(x) \leq 4\overline{\Phi}(x)^2$. With $a = \sqrt{\kappa}$

$$\mathbb{E}\{\exp(aM_1)\} = 1 + a\int_0^\infty \exp(ax)h(x)\,\mathrm{d}x \sim \frac{2\mathcal{H}_2}{\sqrt{\pi}a}\exp(a^2/4).$$

To justify the last step, split the integral at 1 and $a/4$. The first two parts are negligible compared with $a^{-1}\exp(a^2/4)$, using $h \le 1$ and $h(x) \le Cx^{-2}\exp(-x^2)$ for $x \ge 1$. On $x \ge a/4$, the tail equivalence is uniform; setting $x = a/2+y$ reduces the normalised integral to Gaussian dominated convergence, since $a^2/(a/2+y)^2 \le 16$ for $y \ge -a/4$. As $\mathbb{E}\{\exp(-aM_1)\} \le 1$, the identity

$$C_\kappa = \exp(-\kappa/2)(\mathbb{E}\{\exp(\sqrt{\kappa}M_1)\} + \mathbb{E}\{\exp(-\sqrt{\kappa}M_1)\})$$

yields the first relation in (3.17). Finally, $2\overline{\Phi}(\sqrt{\kappa/2}) \sim 2\exp(-\kappa/4)/\sqrt{\pi\kappa}$, proving the ratio limit. □

The coefficient C_κ is a finite-window Brownian expectation, whereas the terminal coefficient in (3.14) is an ordinary normal tail. The small-κ limit recovers a single Brownian branch, and the large-κ limit of the terminal-tail ratio recovers the two-coordinate constant $\mathcal{H}_2$. The argument uses Corollary 2.2 only for the fixed random variable (rv) M_1; it does not apply that result directly to a degenerating covariance family.

Proof of Theorem 2.3. First suppose that $\kappa = u^2\delta \to \infty$, and write $r = 1-\delta$. The terminal covariance is

$$\Sigma_\delta = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix},$$

with eigenvalues δ and $2-\delta$. Let f_δ denote its Gaussian density. Substituting $\boldsymbol{y} = u\mathbf{1}_2 + (2-\delta)\boldsymbol{z}/u$ in the terminal orthant integral yields

$$p^{\mathrm{sh}}(u) = \frac{(2-\delta)^2}{u^2} f_\delta(u\mathbf{1}_2) \int_{[0,\infty)^2} \exp(-z_1-z_2)\exp\left(-\frac{(2-\delta)^2}{2u^2}\boldsymbol{z}^\top\Sigma_\delta^{-1}\boldsymbol{z}\right)\mathrm{d}\boldsymbol{z}.$$

The second exponential is at most one and tends to one for each fixed $\boldsymbol{z}$, since $u^{-2}\|\Sigma_\delta^{-1}\| = 1/\kappa \to 0$. By dominated convergence, the integral tends to one. Evaluating $f_\delta(u\mathbf{1}_2)$ proves the terminal asymptotic in (2.13).
For the path probability, let $X \sim N(0,r)$ be the common value at the split, and set

$$V_u = \frac{u-X}{\sqrt{\delta}}, \qquad h(v) = \mathbb{P}\{M_1 > v\}, \qquad q(v) = \overline{\Phi}(v)^2.$$

Conditioning on X and scaling the independent terminal increments yields

$$A_u := \mathbb{E}\{h(V_u)\} \le \Psi^{\mathrm{sh}}(u) \le A_u + 2\overline{\Phi}(u/\sqrt{r}), \qquad p^{\mathrm{sh}}(u) = \mathbb{E}\{q(V_u)\}.$$

The added term bounds all crossings before the split. The fixed independent-leaf case of Corollary 2.2 yields

$$\frac{h(v)}{q(v)} \longrightarrow \mathcal{H}_2, \qquad v \to \infty.$$

Also $h(v) \le 4q(v)$ for $v \ge 0$, by independence and reflection. For $v < 0$, $h(v) = 1$ and $q(v) \ge 1/4$, so $0 \le h(v)/q(v) \le 4$ on the whole real line.
Define a probability measure on the values of V_u by

$$\mathbb{Q}_u\{V_u \in H\} = \frac{\mathbb{E}\{q(V_u)\mathbb{I}(V_u \in H)\}}{p^{\mathrm{sh}}(u)}.$$

For each fixed $K \ge 0$, Mills' ratio and the terminal asymptotic yield

$$\begin{aligned}\mathbb{Q}_u\{V_u \le K\} &\le \frac{\mathbb{P}\left\{X \ge u - K\sqrt{\delta}\right\}}{p^{\mathrm{sh}}(u)} \\ &\le C_K\sqrt{\kappa}\exp\left(-\frac{\kappa}{2r(2-\delta)} + \frac{K\sqrt{\kappa}}{r}\right) \longrightarrow 0.\end{aligned}$$

Indeed, for all large u, $r \geq 1/2$, and the last exponent is at most $-\kappa/4+2K\sqrt{\kappa}$. Thus $V_u \to \infty$ in $\mathbb{Q}_u$-probability. The bounded ratio h/q implies

$$\frac{A_u}{p^{\mathrm{sh}}(u)} = \mathbb{E}_{\mathbb{Q}_u}\left\{\frac{h(V_u)}{q(V_u)}\right\} \longrightarrow \mathcal{H}_2.$$

The same Gaussian tail estimate with $K = 0$ shows that $2\overline{\Phi}(u/\sqrt{r}) = o(p^{\mathrm{sh}}(u))$. This proves (2.13) in full.

Next suppose that $\kappa(u) \to \kappa_0 \in [0, \infty)$. The reflection integral in (3.16) is valid with $\kappa = \kappa(u)$. Its normalised integrand is uniformly dominated on each bounded interval $0 < \kappa \leq K_0$ by $C\exp(2z)\overline{\Phi}(z/\sqrt{K_0})^2$. Brownian scaling yields $M_{\kappa(u)} \stackrel{d}{=} \sqrt{\kappa(u)}M_1$, so its tail converges for almost every $z > 0$. The same dominated convergence argument as in Lemma 3.12 therefore yields

$$\frac{\Psi^{\mathrm{sh}}(u)}{\overline{\Phi}(u)} \to C_{\kappa_0}, \qquad \frac{p^{\mathrm{sh}}(u)}{\overline{\Phi}(u)} \to D_{\kappa_0}.$$

For the terminal probability the normalised conditional integrand remains bounded by one, also when $\kappa_0 = 0$. The functions C_κ, D_κ are continuous and strictly positive on $[0, \infty)$; continuity of C_κ follows from Brownian scaling and Gaussian exponential moments. Since $F(u, \delta) \to 1$ in this bounded-κ case, (2.11) follows.

When $\kappa \to \infty$, combine (2.13), Lemma 3.13, and Mills' ratio, using

$$\frac{u^2}{2-\delta} = \frac{u^2}{2} + \frac{\kappa}{4} + \frac{u^2\delta^2}{4(2-\delta)}.$$

This proves (2.11) in the remaining regime. From every sequence $u \to \infty$, a further subsequence has $\kappa(u)$ tending either to a finite limit or to infinity. The subsequence criterion proves the assertions for arbitrary $\delta(u) \to 0$, including oscillating $\kappa(u)$. □

We next turn to exceedance by at least one leaf. Define the probability

$$\Psi^{\vee}_{\mathcal{T}}(u) = \mathbb{P}\{\exists t \in [0, T],\ \exists i \in \mathcal{L} : B_i(t) - c_i t + x_i > q_i u\}.$$

For a single leaf put

$$\psi_i(u) = \mathbb{P}\{\exists t \in [0, T] : B_i(t) - c_i t + x_i > q_i u\}.$$

Proposition 3.14. *For every finite Brownian metric tree with positive terminal edge lengths*

$$\Psi^{\vee}_{\mathcal{T}}(u) \sim \sum_{i=1}^{d} \psi_i(u).$$

Moreover, with

$$a_i(u) = q_i u - x_i, \qquad \overline{\Phi}(y) = \mathbb{P}\{\xi > y\}, \quad \xi \sim N(0, 1),$$

we have the exact identity, for all sufficiently large u

$$\psi_i(u) = \overline{\Phi}\left(\frac{a_i(u) + c_i T}{\sqrt{T}}\right) + \exp(-2c_i a_i(u))\overline{\Phi}\left(\frac{a_i(u) - c_i T}{\sqrt{T}}\right),$$

and hence

$$\begin{aligned}\psi_i(u) &\sim \frac{\sqrt{T}}{\sqrt{2\pi}}\left(\frac{1}{a_i(u) + c_i T} + \frac{1}{a_i(u) - c_i T}\right)\exp\left(-\frac{(a_i(u) + c_i T)^2}{2T}\right)\\ &\sim \sqrt{\frac{2}{\pi}}\frac{\sqrt{T}}{q_i u + c_i T - x_i}\exp\left(-\frac{(q_i u + c_i T - x_i)^2}{2T}\right).\end{aligned}$$

Consequently, in the common-barrier case $q_i = 1$, $c_i = c$, $x_i = 0$

$$\mathbb{P}\{\exists t \in [0, T],\ \exists i \in \mathcal{L} : B_i(t) - ct > u\} \sim d\sqrt{\frac{2}{\pi}}\frac{\sqrt{T}}{u + cT}\exp\left(-\frac{(u + cT)^2}{2T}\right).$$

For a level-synchronous Brownian decision tree with N_k children per particle in generation k, let $P_\eta = \prod_{k=1}^{\eta} N_k = d$. The common-barrier specialisation recovers [2, Theorem 3.1].
The proof is given in Section A.
For the largest branch distance, assume $d \geq 2$. For ordered pairs $i \neq j$, define

$$D_{ij}(t) = B_i(t) - B_j(t), \qquad \ell_{ij} = T - r_{ij} = T - t(i \wedge j).$$

Let

$$\ell_* = \max_{1 \leq i,j \leq d:\ i \neq j} (\ell_{ij}), \qquad \mathcal{P}_* = \{(i,j) : i \neq j,\ \ell_{ij} = \ell_*\}.$$

Thus ℓ_* is the longest post-separation time among ordered pairs of leaves.

Proposition 3.15. *As $u \to \infty$*

$$\mathbb{P}\{\exists t \in [0,T],\ \exists i,j \in \mathcal{L} : B_i(t) - B_j(t) > u\} \sim |\mathcal{P}_*| \frac{2\sqrt{\ell_*}}{u\sqrt{\pi}} \exp\left(-\frac{u^2}{4\ell_*}\right).$$

Equivalently, the same asymptotic holds for $\mathbb{P}\Big\{\sup_{t\in[0,T]} \max_{1\leq i,j\leq d}(|B_i(t) - B_j(t)|) > u\Big\}$, because ordered pairs include both orientations.

The proof is given in Section A.
Suppress any initial unary generations. For a level-synchronous Brownian decision tree with first genuine branching time τ_1, the maximal post-separation length is

$$\ell_* = T - \tau_1.$$

If the first generation has children with leaf counts $n_1, \ldots, n_{N_1}$, then the dominant ordered pairs are exactly those whose two leaves lie in different first-generation subtrees, and hence

$$|\mathcal{P}_*| = \sum_{1 \leq a,b \leq N_1 : a \neq b} n_a n_b = P_\eta^2 - \sum_{a=1}^{N_1} n_a^2, \qquad P_\eta = \sum_{a=1}^{N_1} n_a.$$

In the balanced case $n_a = P_\eta / N_1$ this becomes

$$|\mathcal{P}_*| = N_1(N_1 - 1)\left(\frac{P_\eta}{N_1}\right)^2 = P_\eta^2 \frac{N_1 - 1}{N_1}.$$

Thus Proposition 3.15 recovers the Brownian decision-tree diameter asymptotic of [2, Theorem 3.2].

Appendix A. Auxiliary results

Lemma A.1. *Let $d \geq 1$, let Σ be a symmetric positive definite $d \times d$ matrix, and let $\boldsymbol{b} \in \mathbb{R}^d \setminus (-\infty, 0]^d$. The problem*

$$\Pi_\Sigma(\boldsymbol{b}): \qquad \text{minimise } \boldsymbol{x}^\top \Sigma^{-1} \boldsymbol{x} \quad \text{subject to } \boldsymbol{x} \geq \boldsymbol{b}$$

has a unique minimiser $\widetilde{\boldsymbol{b}}$. There is a unique non-empty set $I \subset \{1, \ldots, d\}$ such that, with $J = I^c$

$$\widetilde{\boldsymbol{b}}_I = \boldsymbol{b}_I \neq \boldsymbol{0}_I, \qquad \widetilde{\boldsymbol{b}}_J = \Sigma_{JI}\Sigma_{II}^{-1}\boldsymbol{b}_I \geq \boldsymbol{b}_J, \qquad \Sigma_{II}^{-1}\boldsymbol{b}_I > \boldsymbol{0}_I.$$

Moreover, for every $\boldsymbol{x} \in \mathbb{R}^d$

$$\boldsymbol{x}^\top \Sigma^{-1} \widetilde{\boldsymbol{b}} = \boldsymbol{x}_I^\top \Sigma_{II}^{-1} \boldsymbol{b}_I,$$

and

$$\min_{\boldsymbol{x} \geq \boldsymbol{b}} (\boldsymbol{x}^\top \Sigma^{-1} \boldsymbol{x}) = \widetilde{\boldsymbol{b}}^\top \Sigma^{-1} \widetilde{\boldsymbol{b}} = \boldsymbol{b}_I^\top \Sigma_{II}^{-1} \boldsymbol{b}_I > 0.$$

If $\boldsymbol{w} = \Sigma^{-1}\widetilde{\boldsymbol{b}}$, then

$$\boldsymbol{w}_I = \Sigma_{II}^{-1}\boldsymbol{b}_I > \boldsymbol{0}_I, \qquad \boldsymbol{w}_J = \boldsymbol{0}_J.$$

Proof. The quadratic form is coercive and the feasible set is closed and nonempty, so the minimum is attained. Strict convexity ensures uniqueness. The KKT conditions yield a unique vector $\boldsymbol{w} = \Sigma^{-1}\widetilde{\boldsymbol{b}}$ such that

$$\boldsymbol{w} \geq \mathbf{0}, \qquad \widetilde{\boldsymbol{b}} \geq \boldsymbol{b}, \qquad w_i(\widetilde{b}_i - b_i) = 0, \quad i = 1, \ldots, d.$$

Let $I = \{i \in \{1, \ldots, d\} : w_i > 0\}$. If $I = \varnothing$, then $\widetilde{\boldsymbol{b}} = 0$, which would imply $\boldsymbol{b} \leq 0$, contrary to the assumption. Hence $I \neq \varnothing$. Since $\widetilde{\boldsymbol{b}} = \Sigma \boldsymbol{w}$ and $\boldsymbol{w}_J = 0$, the active equations yield

$$\boldsymbol{b}_I = \widetilde{\boldsymbol{b}}_I = \Sigma_{II} \boldsymbol{w}_I,$$

so $\boldsymbol{w}_I = \Sigma_{II}^{-1} \boldsymbol{b}_I > 0$, and then

$$\widetilde{\boldsymbol{b}}_J = \Sigma_{JI} \Sigma_{II}^{-1} \boldsymbol{b}_I \geq \boldsymbol{b}_J.$$

The remaining identities follow from $\Sigma^{-1}\widetilde{\boldsymbol{b}} = \boldsymbol{w}$. The uniqueness of I follows from that of $\boldsymbol{w}$. □

Lemma A.2. *Let $K \subset \mathbb{R}^m$ be compact, and let $X(t)$, $t \in K$, be a centred GRF with continuous sample paths. Suppose that*

$$\sigma^2 = \sup_{t \in K} \operatorname{Var} X(t) > 0$$

and that, for some constants $C_0 < \infty$ and $\beta > 0$

$$\mathbb{E}\big\{(X(t) - X(s))^2\big\} \leq C_0 \|t - s\|^{\beta}, \qquad s, t \in K.$$

Then there exist constants $C < \infty$ and $\alpha < \infty$ such that, for all sufficiently large u

$$\mathbb{P}\left\{\sup_{t \in K} X(t) > u\right\} \leq C u^{\alpha} \exp\left(-\frac{u^2}{2\sigma^2}\right).$$

Proof. This is the standard Piterbarg inequality for GRFs on compact parameter sets with polynomial metric entropy; see [5, Theorem 8.1]. The increment condition implies the required entropy bound for the canonical pseudometric

$$d_X(s,t) = \big(\mathbb{E}\big\{(X(t) - X(s))^2\big\}\big)^{1/2}.$$

□

Lemma A.3. *As $u \to \infty$*

$$p_T(u) \sim \pi_{\mathcal{J}}(\boldsymbol{b}_{\mathcal{J}}) \frac{u^{-|\mathcal{I}|}}{(2\pi)^{|\mathcal{I}|/2} |\Sigma_{\mathcal{I}\mathcal{I}}|^{1/2} \prod_{i \in \mathcal{I}} \lambda_i} \times \exp\left(-\frac{1}{2}(u\boldsymbol{q}_{\mathcal{I}} + \boldsymbol{a}_{\mathcal{I}})^{\top} \Sigma_{\mathcal{I}\mathcal{I}}^{-1} (u\boldsymbol{q}_{\mathcal{I}} + \boldsymbol{a}_{\mathcal{I}})\right).$$

Proof. Put $m = |\mathcal{I}|$, $A = \Sigma_{\mathcal{I}\mathcal{I}}$ and $\boldsymbol{y}_u = u\boldsymbol{q}_{\mathcal{I}} + \boldsymbol{a}_{\mathcal{I}}$, and let ϕ_A be the density of $N(\mathbf{0}, A)$. For $\boldsymbol{z} \geq \mathbf{0}$, let $g_u(\boldsymbol{z})$ be the conditional probability of the inactive terminal inequalities given $\boldsymbol{Z}_{\mathcal{I}} = \boldsymbol{y}_u + \boldsymbol{z}/u$, with $g_u = 1$ if $\mathcal{J}$ is empty. Integrating over the active coordinates yields the exact identity

$$\frac{p_T(u)}{u^{-m}\phi_A(\boldsymbol{y}_u)} = \int_{[0,\infty)^m} g_u(\boldsymbol{z}) \exp\left(-\boldsymbol{z}^{\top} A^{-1} \boldsymbol{y}_u / u - \frac{\boldsymbol{z}^{\top} A^{-1} \boldsymbol{z}}{2u^2}\right) \mathrm{d}\boldsymbol{z}.$$

For fixed $\boldsymbol{z}$, the conditional inactive mean is $\Sigma_{\mathcal{J}\mathcal{I}} A^{-1}(\boldsymbol{y}_u + \boldsymbol{z}/u)$ when $\mathcal{J} \neq \varnothing$. Subtracting this mean from the inactive threshold yields the limit $\boldsymbol{b}_{\mathcal{J}}$. The conditional covariance is the positive definite Schur complement of A, and hence finite boundary coordinates have no mass at equality and coordinates with limit $-\infty$ impose no restriction. Hence $g_u(\boldsymbol{z}) \to \pi_{\mathcal{J}}(\boldsymbol{b}_{\mathcal{J}})$. Since $A^{-1}\boldsymbol{y}_u/u \to \boldsymbol{\lambda}_{\mathcal{I}} > \mathbf{0}$, the integrand is bounded by $\exp(-\boldsymbol{\lambda}_{\mathcal{I}}^{\top}\boldsymbol{z}/2)$ for all sufficiently large u. Dominated convergence yields the limit $\pi_{\mathcal{J}}(\boldsymbol{b}_{\mathcal{J}}) / \prod_{i \in \mathcal{I}} \lambda_i$, as asserted. □

Lemma A.4. *Let $m = |\mathcal{I}|$ and $\boldsymbol{\lambda}_{\mathcal{I}} \in (0,\infty)^m$. Then*

$$1 \le \mathcal{H}_{\mathcal{I}}(\boldsymbol{\lambda}_{\mathcal{I}}) \le 2^m.$$

For $m = 1$, $\mathcal{H}_{\mathcal{I}}(\boldsymbol{\lambda}_{\mathcal{I}}) = 2$. Moreover, for every $a > 0$

$$\mathcal{H}_{\mathcal{I}}(a\boldsymbol{\lambda}_{\mathcal{I}}) = \mathcal{H}_{\mathcal{I}}(\boldsymbol{\lambda}_{\mathcal{I}}), \qquad H_{\mathcal{I}}(L; a\boldsymbol{\lambda}_{\mathcal{I}}) = H_{\mathcal{I}}(a^2 L; \boldsymbol{\lambda}_{\mathcal{I}}).$$

Proof. Let $X_i(t) = \lambda_i W_i(t) - \lambda_i^2 t$. If $z_i < 0$, then $\sup_{t\ge 0} X_i(t) > z_i$ already at $t = 0$; if $z_i \ge 0$, the classical hitting formula yields

$$\mathbb{P}\Big\{\sup_{t\ge 0} X_i(t) > z_i\Big\} = \exp(-2z_i).$$

The lower bound follows by integrating over $(-\infty, 0)^m$. For the upper bound

$$\{\exists t \ge 0 : X_i(t) > z_i,\ i = 1,\dots,m\} \subset \bigcap_{i=1}^m \{\sup_{t\ge 0} X_i(t) > z_i\},$$

and the coordinates are independent. Therefore

$$\mathcal{H}_{\mathcal{I}}(\boldsymbol{\lambda}_{\mathcal{I}}) \le \prod_{i=1}^m \left(\int_{-\infty}^0 \exp(z_i)\,\mathrm{d}z_i + \int_0^\infty \exp(-2z_i)\exp(z_i)\,\mathrm{d}z_i \right) = 2^m.$$

For $m = 1$, the event inclusion above is an equality and hence the integral equals 2. The scaling identities follow from

$$a\lambda_i W_i(t) - a^2\lambda_i^2 t \overset{d}{=} \lambda_i \widetilde{W}_i(a^2 t) - \lambda_i^2 (a^2 t),$$

then changing variables $r = a^2 t$ and letting $L \to \infty$. □

Lemma A.5. *Let $W_1, \dots, W_m$ be independent standard Brownian motions, and let $\lambda_i > 0$, $i = 1,\dots,m$. For every $L > 0$*

$$\exp\left(-\frac{L}{2}\sum_{i=1}^m \lambda_i^2\right) \int_{\mathbb{R}^m} \mathbb{P}\{\exists s \in [0,L] : \lambda_i W_i(s) > y_i,\ i = 1,\dots,m\} \exp\left(\sum_{i=1}^m y_i\right) \mathrm{d}\boldsymbol{y}$$
$$= \int_{\mathbb{R}^m} \mathbb{P}\big\{\exists s \in [0,L] : \lambda_i W_i(s) - \lambda_i^2 s > y_i,\ i = 1,\dots,m\big\} \exp\left(\sum_{i=1}^m y_i\right) \mathrm{d}\boldsymbol{y}.$$

Proof. For a continuous path $\omega = (\omega_1, \dots, \omega_m)$ on $[0, L]$, define

$$F(\omega) = \int_{\mathbb{R}^m} \mathbb{I}(\exists s \in [0,L] : \lambda_i \omega_i(s) > y_i,\ i = 1,\dots,m) \exp\left(\sum_{i=1}^m y_i\right) \mathrm{d}\boldsymbol{y}.$$

The functional is non-negative and finite almost surely for Brownian paths. For example, it is bounded above by $\exp(\sum_{i=1}^m \lambda_i \sup_{0\le s\le L} \omega_i(s))$. The right-hand side in the displayed identity is

$$\mathbb{E}\{F(W_1 - \lambda_1 \cdot, \dots, W_m - \lambda_m \cdot)\}.$$

By the Cameron–Martin formula on $[0, L]$, this equals

$$\exp\left(-\frac{L}{2}\sum_{i=1}^m \lambda_i^2\right) \mathbb{E}\left\{F(W) \exp\left(-\sum_{i=1}^m \lambda_i W_i(L)\right)\right\}.$$

It remains to remove the terminal exponential factor. Inside the expectation, change variables $r_i = y_i - \lambda_i W_i(L)$. Then

$$F(W) \exp\left(-\sum_{i=1}^m \lambda_i W_i(L)\right)$$
$$= \int_{\mathbb{R}^m} \mathbb{I}(\exists s \in [0,L] : \lambda_i (W_i(s) - W_i(L)) > r_i,\ i = 1,\dots,m) \exp\left(\sum_{i=1}^m r_i\right) \mathrm{d}\boldsymbol{r}.$$

The reversed processes $W_i(L-s)-W_i(L),\ 0\le s\le L$, for $1\le i\le m$ have the joint law of independent standard Brownian motions. Thus

$$\mathbb{E}\left\{F(W)\exp\left(-\sum_{i=1}^m \lambda_i W_i(L)\right)\right\}=\mathbb{E}\{F(W)\}.$$

Multiplying by $\exp\bigl(-L\sum_{i=1}^m \lambda_i^2/2\bigr)$ yields the claimed identity. □

Lemma A.6. *For every fixed $L>0$*

$$\begin{aligned}&\mathbb{P}\bigl\{\exists t\in[T-Lu^{-2},T]: B_i(t)-c_it+x_i>q_iu,\ i=1,\dots,d\bigr\}\\&\quad\sim H_{\mathcal{I}}(L;\boldsymbol{\lambda}_{\mathcal{I}})p_T(u).\end{aligned}$$

Proof. Set $S_u=T-Lu^{-2}$, $\Delta_u=Lu^{-2}$

$$A=\Sigma_{\mathcal{I}\mathcal{I}},\qquad A_u=A-\Delta_u I_{|\mathcal{I}|},\qquad m=|\mathcal{I}|.$$

For large u, $S_u\in[T-\delta_0,T]$, so no branching occurs on $[S_u,T]$, the terminal increments over this interval are independent over leaves, and $\mathrm{Cov}(\boldsymbol{B}_{\mathcal{I}}(S_u))=A_u$. Let f_u and f be the densities of $N(0,A_u)$ and $N(0,A)$, respectively, and put

$$\boldsymbol{y}_T=u\boldsymbol{q}_{\mathcal{I}}+\boldsymbol{a}_{\mathcal{I}},\qquad \boldsymbol{y}_u(\boldsymbol{z})=u\boldsymbol{q}_{\mathcal{I}}+\boldsymbol{a}_{\mathcal{I}}(S_u)-\frac{\boldsymbol{z}}{u}.$$

Conditioning on $\boldsymbol{B}_{\mathcal{I}}(S_u)=\boldsymbol{y}_u(\boldsymbol{z})$ and using the change of variables $\boldsymbol{B}_{\mathcal{I}}(S_u)=\boldsymbol{y}_u(\boldsymbol{z})$ yields

$$M_u(L)=u^{-m}\int_{\mathbb{R}^m}R_u(\boldsymbol{z})f_u(\boldsymbol{y}_u(\boldsymbol{z}))\,\mathrm{d}\boldsymbol{z},$$

where $M_u(L)$ denotes the left-hand probability and $R_u(\boldsymbol{z})$ is the corresponding conditional crossing probability on $[S_u,T]$.
For $t=S_u+s/u^2$, $s\in[0,L]$, the active inequalities are equivalent to

$$u(B_i(S_u+s/u^2)-B_i(S_u))>z_i+c_is/u,\qquad i\in\mathcal{I}.$$

By Brownian scaling, the active future increments $u(B_i(S_u+s/u^2)-B_i(S_u))$, $0\le s\le L$, are independent standard Brownian motions. Hence, at every continuity point of the monotone function

$$J_L(\boldsymbol{z})=\mathbb{P}\{\exists s\in[0,L]: W_i(s)>z_i,\ i\in\mathcal{I}\},$$

the active part converges to $J_L(\boldsymbol{z})$.
It remains to identify the inactive factor inside $R_u(\boldsymbol{z})$. If $\mathcal{J}=\varnothing$, the inactive factor is $\pi_{\mathcal{J}}=1$. Otherwise, given $\boldsymbol{B}_{\mathcal{I}}(S_u)=\boldsymbol{y}_u(\boldsymbol{z})$, write

$$\boldsymbol{V}_u=\boldsymbol{B}_{\mathcal{J}}(S_u)-\Sigma_{\mathcal{J}\mathcal{I}}A_u^{-1}\boldsymbol{y}_u(\boldsymbol{z}).$$

Then $\boldsymbol{V}_u$ is centred Gaussian with covariance

$$C_u=\Sigma_{\mathcal{J}\mathcal{J}}-\Delta_uI_{|\mathcal{J}|}-\Sigma_{\mathcal{J}\mathcal{I}}A_u^{-1}\Sigma_{\mathcal{I}\mathcal{J}}\longrightarrow C,$$

where C is the Schur complement of A in Σ. The inactive threshold at time S_u, centred by the conditional mean, is

$$\begin{aligned}d_u(\boldsymbol{z})&:=u\boldsymbol{q}_{\mathcal{J}}+\boldsymbol{a}_{\mathcal{J}}(S_u)-\Sigma_{\mathcal{J}\mathcal{I}}A_u^{-1}\boldsymbol{y}_u(\boldsymbol{z})\\&=u\bigl(\boldsymbol{q}_{\mathcal{J}}-\Sigma_{\mathcal{J}\mathcal{I}}A^{-1}\boldsymbol{q}_{\mathcal{I}}\bigr)+\boldsymbol{a}_{\mathcal{J}}-\Sigma_{\mathcal{J}\mathcal{I}}A^{-1}\boldsymbol{a}_{\mathcal{I}}+o(1),\end{aligned}$$

uniformly for $\boldsymbol{z}$ in compact sets. Thus $d_u(\boldsymbol{z})\to\boldsymbol{b}_{\mathcal{J}}$, with the strictly inactive coordinates tending to $-\infty$.
For $\eta>0$, let $\boldsymbol{b}_{\mathcal{J}}\pm\eta$ mean that $\pm\eta$ is added only to those coordinates of $\boldsymbol{b}_{\mathcal{J}}$ that are finite; coordinates equal to $-\infty$ remain equal to $-\infty$. Since the inactive future increments satisfy

$$\sup_{0\le s\le L/u^2}\max_{j\in\mathcal{J}}(|B_j(S_u+s)-B_j(S_u)|)=O_{\mathbb{P}}(u^{-1})$$

and the deterministic trends change by $O(u^{-2})$, the inactive constraints can be bounded by their time-S_u versions. More precisely, for every continuity point of J_L

$$\pi_{\mathcal{J}}(\boldsymbol{b}_{\mathcal{J}}+\eta)J_L(\boldsymbol{z}) \le \liminf_{u\to\infty} R_u(\boldsymbol{z}) \le \limsup_{u\to\infty} R_u(\boldsymbol{z}) \le \pi_{\mathcal{J}}(\boldsymbol{b}_{\mathcal{J}}-\eta)J_L(\boldsymbol{z}).$$

Indeed, the lower bound follows by requiring $\boldsymbol{V}_u > d_u(\boldsymbol{z})+\eta$ and by using the fact that the inactive future increments are uniformly smaller than η with probability tending to one; the upper bound follows because a full crossing implies either $\boldsymbol{V}_u > d_u(\boldsymbol{z})-\eta$ or such a fluctuation of the inactive future increments occurs. The active future increments, inactive future increments, and $\boldsymbol{V}_u$ are independent on the terminal window. Letting $\eta \downarrow 0$ and using continuity of the non-degenerate conditional Gaussian law on the finite inactive coordinates yields

$$R_u(\boldsymbol{z}) \longrightarrow \pi_{\mathcal{J}}(\boldsymbol{b}_{\mathcal{J}})J_L(\boldsymbol{z})$$

for Lebesgue-a.e. $\boldsymbol{z}$.
Next

$$A_u^{-1} = A^{-1} + \Delta_u A^{-2} + O(\Delta_u^2), \qquad \boldsymbol{y}_u(\boldsymbol{z}) = \boldsymbol{y}_T - \frac{\boldsymbol{z}}{u} - \Delta_u \boldsymbol{c}_{\mathcal{I}},$$

and therefore, for fixed $\boldsymbol{z}$

$$\frac{f_u(\boldsymbol{y}_u(\boldsymbol{z}))}{f(\boldsymbol{y}_T)} \longrightarrow \exp\left(\boldsymbol{z}^\top \boldsymbol{\lambda}_{\mathcal{I}} - \frac{L}{2}\boldsymbol{\lambda}_{\mathcal{I}}^\top \boldsymbol{\lambda}_{\mathcal{I}}\right).$$

We now justify dominated convergence for the integrand $R_u(\boldsymbol{z})f_u(\boldsymbol{y}_u(\boldsymbol{z}))/f(\boldsymbol{y}_T)$. Fix $0<\varepsilon<\min_{i\in\mathcal{I}}(\lambda_i)$. For all large u

$$\frac{f_u(\boldsymbol{y}_u(\boldsymbol{z}))}{f(\boldsymbol{y}_T)} \le C_L \exp\left(\sum_{i\in\mathcal{I}} (\lambda_i + \varepsilon \operatorname{sign} z_i) z_i\right).$$

This bound is integrable in the negative coordinate directions. To control the positive directions, set

$$F(\boldsymbol{z}) = \{i \in \mathcal{I} : z_i > 1\}, \qquad Z_F = \sum_{i\in F(\boldsymbol{z})} z_i.$$

If $F(\boldsymbol{z}) \neq \varnothing$, then the active part of the crossing event implies

$$\sup_{0\le s\le L} \sum_{i\in F(\boldsymbol{z})} W_i(s) > Z_F - C_L$$

for all large u. Since $\sum_{i\in F} W_i$ is a Brownian motion with variance parameter $|F| \le m$, the reflection principle yields

$$R_u(\boldsymbol{z}) \le C_L \exp\left(-\frac{((Z_F - C_L)_+)^2}{2mL}\right),$$

with the right-hand side interpreted as C_L when $F=\varnothing$. On each of the finitely many regions determined by $F(\boldsymbol{z})$, the Gaussian decay in Z_F dominates the linear exponential factor in the positive coordinates, while the negative coordinates are controlled by $\exp(\sum_{i\in\mathcal{I}:z_i<0}(\lambda_i-\varepsilon)z_i)$. Hence the integrand is bounded by an integrable function depending on L, but not on u. Consequently we have

$$M_u(L) \sim u^{-m} f(\boldsymbol{y}_T)\pi_{\mathcal{J}}(\boldsymbol{b}_{\mathcal{J}}) \exp\left(-\frac{L}{2}\boldsymbol{\lambda}_{\mathcal{I}}^\top \boldsymbol{\lambda}_{\mathcal{I}}\right) \int_{\mathbb{R}^m} J_L(\boldsymbol{z}) \exp(\boldsymbol{z}^\top \boldsymbol{\lambda}_{\mathcal{I}})\, \mathrm{d}\boldsymbol{z}.$$

By Lemma A.3

$$p_T(u) \sim \pi_{\mathcal{J}}(\boldsymbol{b}_{\mathcal{J}}) \frac{u^{-m} f(\boldsymbol{y}_T)}{\prod_{i\in\mathcal{I}} \lambda_i}.$$

After the change of variables $y_i = \lambda_i z_i$, the ratio $M_u(L)/p_T(u)$ converges to

$$\exp\left(-\frac{L}{2}\|\boldsymbol{\lambda}_{\mathcal{I}}\|^2\right) \int_{\mathbb{R}^m} \mathbb{P}\{\exists s \in [0,L] : \lambda_i W_i(s) > y_i,\ i\in\mathcal{I}\} \exp\left(\sum_{i\in\mathcal{I}} y_i\right) \mathrm{d}\boldsymbol{y}.$$

By the Brownian exponential-shift identity, Lemma A.5, the last display equals $H_{\mathcal{I}}(L;\boldsymbol{\lambda}_{\mathcal{I}})$. This proves the lemma. □

The following stopping-time argument adapts the terminal-tail comparison of [4] to the edge-Brownian construction.

Lemma A.7. *Let $A \subset \{1,\dots,d\}$ be non-empty and let $S \in (0,T]$. Define*

$$\Psi_A(u;S) = \mathbb{P}\{\exists t \in [0,S] : B_i(t) - c_i t + x_i > q_i u,\ i \in A\},$$

and

$$p_A(u;S) = \mathbb{P}\{B_i(S) - c_i S + x_i > q_i u,\ i \in A\}.$$

Then, for all $u > 0$

$$\Psi_A(u;S) \le C_A(S) p_A(u;S),$$

where one may take

$$C_A(S) = 2^{|\{i\in A: c_i \le 0\}|} \prod_{\substack{i\in A\\ c_i>0}} \left(\mathbb{P}\Big\{\xi > c_i\sqrt{S}\Big\}\right)^{-1}, \qquad \xi \sim N(0,1).$$

Proof. Let

$$X_i(t) = B_i(t) - c_i t + x_i - q_i u, \qquad Y_A(t) = \min_{i\in A}(X_i(t)).$$

For $\varepsilon > 0$, define

$$\tau_\varepsilon = \inf\{t \in [0,S] : Y_A(t) \ge \varepsilon\}, \qquad \inf \varnothing = \infty.$$

Since Y_A is continuous and adapted, τ_ε is a stopping time. Moreover

$$\{\tau_\varepsilon \le S\} = \left\{\sup_{0\le t\le S} Y_A(t) \ge \varepsilon\right\} \uparrow \left\{\sup_{0\le t\le S} Y_A(t) > 0\right\}$$

as $\varepsilon \downarrow 0$. The last event defines $\Psi_A(u;S)$.
On $\{\tau_\varepsilon \le S\}$, put $r = S - \tau_\varepsilon$. If $r = 0$, the terminal inequalities hold because $X_i(S) \ge \varepsilon$. If $r > 0$, they hold whenever

$$B_i(S) - B_i(\tau_\varepsilon) > c_i r - \varepsilon, \qquad i \in A.$$

By the strong Markov property of the finite edge-Brownian construction, τ_ε is a stopping time for the height filtration and, conditional on $\mathcal{F}_{\tau_\varepsilon}$, the unexplored edge increments above height τ_ε are independent centred Brownian increments. Consequently, the conditional law of

$$(B_i(S) - B_i(\tau_\varepsilon))_{i\in A}$$

is centred Gaussian with covariance $\Gamma_A(S) - \Gamma_A(\tau_\varepsilon)$. Its variances are equal to r, and its covariances are non-negative because each entry $t \mapsto \Gamma_{ij}(t)$ is non-decreasing. On $\{r > 0\}$, Gaussian association [6] yields

$$\begin{aligned}\mathbb{P}\{B_i(S) - B_i(\tau_\varepsilon) > c_i r - \varepsilon,\ i \in A \mid \mathcal{F}_{\tau_\varepsilon}\}\\ \ge \prod_{i\in A} \mathbb{P}\{\xi > c_i\sqrt{r} - \varepsilon/\sqrt{r}\} \ge C_A(S)^{-1}.\end{aligned}$$

The last inequality follows because, for $c_i \le 0$, the displayed threshold is at most 0, while for $c_i > 0$, it is at most $c_i\sqrt{S}$. On $\{r = 0\}$ the conditional probability of the terminal event is 1, which is also at least $C_A(S)^{-1}$. Thus

$$p_A(u;S) \ge C_A(S)^{-1}\mathbb{P}\{\tau_\varepsilon \le S\}.$$

Letting $\varepsilon \downarrow 0$ proves the bound. □

Lemma A.8. *There exist constants $C, c > 0$, independent of u and L, such that, for every fixed $L > 0$*

$$\begin{aligned}\mathbb{P}\big\{\exists t \in [0, T - Lu^{-2}] : B_i(t) - c_i t + x_i > q_i u,\ i = 1,\dots,d\big\}\\ \le C\exp(-cL)p_T(u) + o(p_T(u)).\end{aligned}$$

Proof. Let $S_u = T - Lu^{-2}$. For every fixed $L > 0$, $S_u > 0$ and $S_u \in [T - \delta_0, T]$ for all large u. The full exceedance event on $[0, S_u]$ implies the corresponding active-coordinate event. By Lemma A.7, and uniformly over $S_u \le T$

$$\begin{aligned}\mathbb{P}\{\exists t \in [0, S_u] : B_i(t) - c_i t + x_i > q_i u,\ i \in \mathcal{I}\}\\ \le C_0 \mathbb{P}\{\boldsymbol{B}_{\mathcal{I}}(S_u) > u\boldsymbol{q}_{\mathcal{I}} + \boldsymbol{a}_{\mathcal{I}}(S_u)\}.\end{aligned}$$

Here C_0 is independent of u and L, because the constant in Lemma A.7 is bounded on $S \in (0, T]$. Indeed, for $c_i > 0$, $\mathbb{P}\left\{\xi > c_i\sqrt{S}\right\} \ge \mathbb{P}\left\{\xi > c_i\sqrt{T}\right\}$, while the coordinates with $c_i \le 0$ contribute only the fixed factor 2.
Put

$$A = \Sigma_{\mathcal{I}\mathcal{I}}, \qquad A_u = A - Lu^{-2} I_{|\mathcal{I}|}, \qquad \boldsymbol{y}_u = u\boldsymbol{q}_{\mathcal{I}} + \boldsymbol{a}_{\mathcal{I}} - Lu^{-2}\boldsymbol{c}_{\mathcal{I}}, \qquad \boldsymbol{y}_T = u\boldsymbol{q}_{\mathcal{I}} + \boldsymbol{a}_{\mathcal{I}}.$$

Then $\mathrm{Cov}(\boldsymbol{B}_{\mathcal{I}}(S_u)) = A_u$. Since $\boldsymbol{\lambda}_{\mathcal{I}} = A^{-1}\boldsymbol{q}_{\mathcal{I}} > \boldsymbol{0}$, also $A_u^{-1}\boldsymbol{q}_{\mathcal{I}} > \boldsymbol{0}$ for all large u. The full-active Savage expansion also applies to the varying covariance A_u. Indeed, with $m = |\mathcal{I}|$ and f_{A_u} the density of $N(0, A_u)$

$$\begin{aligned}&\mathbb{P}\{N(0, A_u) > \boldsymbol{y}_u\}\\ &= u^{-m} f_{A_u}(\boldsymbol{y}_u) \int_{[0,\infty)^m} \exp\left(-\boldsymbol{z}^\top A_u^{-1}\boldsymbol{y}_u/u - \frac{\boldsymbol{z}^\top A_u^{-1}\boldsymbol{z}}{2u^2}\right) \mathrm{d}\boldsymbol{z}.\end{aligned}$$

Since $A_u^{-1}\boldsymbol{y}_u/u \to \boldsymbol{\lambda}_{\mathcal{I}} > \boldsymbol{0}$, the integrand is dominated by $\exp(-\frac{1}{2}\boldsymbol{z}^\top\boldsymbol{\lambda}_{\mathcal{I}})$ for all large u. Dominated convergence yields the required tail expansion. Moreover, for $\Delta_u = Lu^{-2}$

$$A_u^{-1} = A^{-1} + \Delta_u A^{-2} + O(\Delta_u^2),$$

where the error is uniform for $0 \le \Delta_u \le \delta$ with any fixed $\delta < \lambda_{\min}(A)$. Consequently, for every fixed L

$$\begin{aligned}\boldsymbol{y}_u^\top A_u^{-1}\boldsymbol{y}_u - \boldsymbol{y}_T^\top A^{-1}\boldsymbol{y}_T &= L\boldsymbol{q}_{\mathcal{I}}^\top A^{-2}\boldsymbol{q}_{\mathcal{I}} + o(1)\\ &= L\|\boldsymbol{\lambda}_{\mathcal{I}}\|^2 + o(1).\end{aligned}$$

The determinant and product terms in the two Savage expansions have ratio $1 + o(1)$. Hence

$$\frac{\mathbb{P}\{\boldsymbol{B}_{\mathcal{I}}(S_u) > u\boldsymbol{q}_{\mathcal{I}} + \boldsymbol{a}_{\mathcal{I}}(S_u)\}}{\mathbb{P}\{\boldsymbol{B}_{\mathcal{I}}(T) > u\boldsymbol{q}_{\mathcal{I}} + \boldsymbol{a}_{\mathcal{I}}\}} \longrightarrow \exp\left(-\frac{L}{2}\|\boldsymbol{\lambda}_{\mathcal{I}}\|^2\right).$$

Equivalently, for any fixed $c < \|\boldsymbol{\lambda}_{\mathcal{I}}\|^2/2$

$$\limsup_{u\to\infty} \frac{\mathbb{P}\{\boldsymbol{B}_{\mathcal{I}}(S_u) > u\boldsymbol{q}_{\mathcal{I}} + \boldsymbol{a}_{\mathcal{I}}(S_u)\}}{\mathbb{P}\{\boldsymbol{B}_{\mathcal{I}}(T) > u\boldsymbol{q}_{\mathcal{I}} + \boldsymbol{a}_{\mathcal{I}}\}} \le \exp(-cL).$$

Here L is fixed as $u \to \infty$, while the constants C_0 and c do not depend on L.
Finally, the active-set terminal tail expansion yields

$$p_T(u) \sim \pi_{\mathcal{J}}(\boldsymbol{b}_{\mathcal{J}})\mathbb{P}\{\boldsymbol{B}_{\mathcal{I}}(T) > u\boldsymbol{q}_{\mathcal{I}} + \boldsymbol{a}_{\mathcal{I}}\},$$

with $\pi_{\mathcal{J}}(\boldsymbol{b}_{\mathcal{J}}) > 0$. Combining the previous displays and enlarging the multiplicative constant proves

$$\begin{aligned}\mathbb{P}\{\exists t \in [0, T - Lu^{-2}] : B_i(t) - c_i t + x_i > q_i u,\ i = 1, \dots, d\}\\ \le C\exp(-cL)p_T(u) + o(p_T(u)),\end{aligned}$$

for every fixed $L > 0$, with C, c independent of u and L. □

Direct endpoint-window proof of Corollary 2.2. Fix $L > 0$ and put $S_u = T - Lu^{-2}$. By Lemma A.6

$$\mathbb{P}\{\exists t \in [S_u, T] : B_i(t) - c_i t + x_i > q_i u,\ i = 1, \dots, d\} \sim H_{\mathcal{I}}(L; \boldsymbol{\lambda}_{\mathcal{I}})p_T(u).$$

Thus

$$\liminf_{u\to\infty} \frac{\Psi_{\mathcal{T}}(u)}{p_T(u)} \ge H_{\mathcal{I}}(L; \boldsymbol{\lambda}_{\mathcal{I}}).$$

Letting $L \to \infty$ yields the lower bound $\mathcal{H}_{\mathcal{I}}(\boldsymbol{\lambda}_{\mathcal{I}})$.

For the upper bound, split $[0,T]=[0,S_u]\cup[S_u,T]$. Lemmas A.6 and A.8 yield

$$\limsup_{u\to\infty}\frac{\Psi_{\mathcal{T}}(u)}{p_T(u)}\le H_{\mathcal{I}}(L;\boldsymbol{\lambda}_{\mathcal{I}})+C\exp(-cL).$$

Let $L\to\infty$. Since $H_{\mathcal{I}}(L;\boldsymbol{\lambda}_{\mathcal{I}})\uparrow\mathcal{H}_{\mathcal{I}}(\boldsymbol{\lambda}_{\mathcal{I}})$ and $\mathcal{H}_{\mathcal{I}}$ is finite by Lemma A.4, the upper and lower bounds coincide. The explicit formula follows from Lemma A.3. □

Remark A.9. *For $\boldsymbol{c}=\boldsymbol{x}=\mathbf{0}$, Corollary 2.2 follows from [1, Theorem 2.4(2) and Remark 2.5]. To check the endpoint-maximality hypothesis when $\Gamma(t)$ may be singular at earlier times, write the generalised variance as*

$$g(t)=\min_{\substack{\boldsymbol{z}\ge\mathbf{0}\\ \boldsymbol{z}^\top\boldsymbol{q}=1}}(\boldsymbol{z}^\top\Gamma(t)\boldsymbol{z}).$$

Since $\boldsymbol{q}>\mathbf{0}$, the set of minimisation is compact and g is continuous. For every $t<T$, the matrix $\Sigma-\Gamma(t)$ is positive definite: it contains the independent residual terminal-edge variances $T-\max(t,t(p(i)))>0$, in addition to a non-negative definite matrix. If $\boldsymbol{z}_T$ minimises the expression at T, then

$$g(t)\le\boldsymbol{z}_T^\top\Gamma(t)\boldsymbol{z}_T<\boldsymbol{z}_T^\top\Sigma\boldsymbol{z}_T=g(T).$$

Thus T is the unique maximiser. A continuous covariance square root exists on the full interval, including times when $\Gamma(t)$ is singular.
Reverse time by setting $\widehat{\boldsymbol{B}}(s)=\boldsymbol{B}(T-s)$. On the terminal edge window its covariance is

$$R(s,t)=\Sigma-\max(s,t)I_d.$$

In the notation of [1], set $A(s)=\Gamma(T-s)^{1/2}$ on $[0,T]$. Near $s=0$ the local parameters are

$$A(s)=(\Sigma-sI_d)^{1/2},\qquad \alpha=\beta=1,$$
$$\Xi=\tfrac12\Sigma^{-1/2},\qquad D=\tfrac12\Sigma^{-1}.$$

The normalised covariance expansion follows from expanding $(\Sigma-tI_d)^{1/2}(\Sigma-sI_d)^{-1/2}$ for $t>s$. Hence $V=ADA^\top=I_d/2$ and $W=\Xi A^\top=I_d/2$, with $A=A(0)$. The positivity parameter of the cited theorem is $\tau_{\boldsymbol{\lambda}}=\frac12\|\boldsymbol{\lambda}\|^2>0$. The required increment bound is immediate from $\mathbb{E}\{\|\boldsymbol{B}(t)-\boldsymbol{B}(s)\|^2\}=d|t-s|$. The resulting Piterbarg constant is $\mathcal{H}_{\mathcal{I}}(\boldsymbol{\lambda}_{\mathcal{I}})$: coordinates with zero multiplier contribute the factor $\int_{-\infty}^0\exp(z)\,\mathrm{d}z=1$. The proof of Corollary 2.2 includes the fixed trends and shifts and explicitly identifies the inactive Gaussian boundary factor.

Reflection verification of the unit-barrier case in Example 3.2. Here ϕ_r denotes the density of $N(0,r)$. Write $r=r_u$. Since $M_{\mathcal{G}}\ge0$ is independent of the trunk, the reflection argument proving (3.15) yields

$$\Psi_{\mathcal{G}}(u)=\mathbb{E}\left\{\overline{\Phi}\left(\frac{u-M_{\mathcal{G}}/u}{\sqrt{r}}\right)+\overline{\Phi}\left(\frac{u+M_{\mathcal{G}}/u}{\sqrt{r}}\right)\right\}.$$

The same argument yields (3.16) with $h_\kappa(y)$ replaced by $h_{\mathcal{G}}(y)=\mathbb{P}\{M_{\mathcal{G}}>y\}$. Every leaf process G_i is a standard Brownian motion, so

$$0\le M_{\mathcal{G}}\le\sup_{0\le s\le\kappa}G_1(s),\qquad h_{\mathcal{G}}(y)\le2\overline{\Phi}(y/\sqrt{\kappa}),\quad y\ge0.$$

This Gaussian bound justifies the same dominated convergence argument and proves the first asymptotic in (3.9).
For the terminal event, condition on $W_0(r)=u+z/u$ to obtain

$$\frac{p_{\mathcal{G}}(u)}{\overline{\Phi}(u)}=\int_{\mathbb{R}}\frac{\phi_r(u+z/u)}{u\overline{\Phi}(u)}\mathbb{P}\{L_{\mathcal{G}}>-z\}\,\mathrm{d}z.$$

The density ratio tends to $\exp(-\kappa/2-z)$. For large u, the integrand is bounded by $C\exp(-z)$ for $z \geq 0$. At $z = -y < 0$, it is bounded by $C\exp(2y)\overline{\Phi}(y/\sqrt{\kappa})$, since $L_{\mathcal{G}} \leq G_1(\kappa)$. Both bounds are integrable. Dominated convergence, followed by Tonelli's theorem, yields

$$\frac{p_{\mathcal{G}}(u)}{\overline{\Phi}(u)} \longrightarrow \exp(-\kappa/2)\int_{\mathbb{R}} \exp(-z)\mathbb{P}\{L_{\mathcal{G}} > -z\}\,\mathrm{d}z = \exp(-\kappa/2)\mathbb{E}\{\exp(L_{\mathcal{G}})\}.$$

The denominator in the ratio limit is positive and finite, because $L_{\mathcal{G}}$ is finite almost surely and $\mathbb{E}\{\exp(L_{\mathcal{G}})\} \leq \mathbb{E}\{\exp(G_1(\kappa))\} = \exp(\kappa/2)$. □

Proof of Proposition 3.14. Let

$$A_i(u) = \{\exists t \in [0,T] : B_i(t) - c_i t + x_i > q_i u\}.$$

The reflection-principle formula for Brownian motion with drift yields the stated identity for $\psi_i(u)$. Mills' ratio then yields the displayed asymptotic expansion.
It remains to prove that intersections are negligible. Put

$$q_* = \min_{1\leq i\leq d}(q_i), \qquad S(u) = \sum_{i=1}^{d}\psi_i(u).$$

Since at least one coordinate has $q_i = q_*$, Mills' ratio implies

$$S(u) \geq C_1 u^{-1}\exp\left(-\frac{(q_*u + C_2)^2}{2T}\right)$$

for some constants $C_1, C_2 > 0$ and all large u. If $\max(q_i, q_j) > q_*$, then

$$\mathbb{P}\{A_i(u) \cap A_j(u)\} \leq \psi_k(u) = o(S(u)),$$

where $k \in \{i, j\}$ is chosen so that $q_k > q_*$.
It remains to consider $q_i = q_j = q_*$ with $i \neq j$. Let $r_{ij} = t(i \wedge j) < T$. Since the deterministic trends and shifts are bounded on $[0,T]$, the event $A_i(u) \cap A_j(u)$ implies

$$\sup_{0\leq s,t\leq T}(B_i(s) + B_j(t)) > 2q_*u - C_{ij}$$

for a constant C_{ij}. For the centred GRF $X_{ij}(s,t) = B_i(s) + B_j(t)$

$$\operatorname{Var} X_{ij}(s,t) = s + t + 2\min(s,t,r_{ij}), \qquad 0 \leq s,t \leq T,$$

and therefore

$$\sigma_{ij}^2 := \sup_{s,t\in[0,T]} \operatorname{Var} X_{ij}(s,t) = 2(T + r_{ij}) < 4T.$$

Moreover

$$\mathbb{E}\big\{(X_{ij}(s,t) - X_{ij}(s',t'))^2\big\} \leq C\big(|s-s'| + |t-t'|\big),$$

so Lemma A.2 applies. It yields constants $C_3 < \infty$ and $\alpha < \infty$ such that

$$\mathbb{P}\{A_i(u) \cap A_j(u)\} \leq C_3 u^{\alpha}\exp\left(-\frac{(2q_*u - C_{ij})^2}{2\sigma_{ij}^2}\right).$$

Since $\sigma_{ij}^2 < 4T$, the exponent above is strictly smaller, on the quadratic scale in u, than the exponent in the lower bound for $S(u)$. Thus $\mathbb{P}\{A_i(u) \cap A_j(u)\} = o(S(u))$. Bonferroni's inequalities and the finiteness of $\mathcal{L}$ now yield

$$S(u) - \sum_{1\leq i<j\leq d}\mathbb{P}\{A_i(u) \cap A_j(u)\} \leq \Psi_T^{\vee}(u) \leq S(u),$$

which proves the proposition. □

Proof of Proposition 3.15. For $i \neq j$, the two leaves coincide up to time r_{ij}. After that time the two descendant increments are independent, and hence

$$\big(D_{ij}(r_{ij}+s)\big)_{0\le s\le \ell_{ij}} \stackrel{d}{=} \big(\sqrt{2}W(s)\big)_{0\le s\le \ell_{ij}},$$

where W is a standard Brownian motion. Hence, for

$$E_{ij}(u) = \{\exists t \in [0,T] : D_{ij}(t) > u\},$$

the reflection principle yields

$$\mathbb{P}\{E_{ij}(u)\} = 2\overline{\Phi}\left(\frac{u}{\sqrt{2\ell_{ij}}}\right) \sim \frac{2\sqrt{\ell_{ij}}}{u\sqrt{\pi}} \exp\left(-\frac{u^2}{4\ell_{ij}}\right).$$

Since the number of ordered pairs is finite

$$\sum_{1\le i,j\le d: i\neq j} \mathbb{P}\{E_{ij}(u)\} \sim |\mathcal{P}_*| \frac{2\sqrt{\ell_*}}{u\sqrt{\pi}} \exp\left(-\frac{u^2}{4\ell_*}\right).$$

It remains to control intersections. If $\ell_{ij} < \ell_*$, then $\mathbb{P}\{E_{ij}(u)\} = o(\chi_*(u))$, where

$$\chi_*(u) = \frac{2\sqrt{\ell_*}}{u\sqrt{\pi}} \exp\left(-\frac{u^2}{4\ell_*}\right).$$

Thus intersections involving a non-dominant pair are negligible.
Consider two distinct dominant ordered pairs $p = (i,j), q = (k,l) \in \mathcal{P}_*$. Set

$$X_{p,q}(s,t) = D_{ij}(s) + D_{kl}(t), \qquad (s,t) \in [0,T]^2.$$

Then $E_{ij}(u) \cap E_{kl}(u)$ implies

$$\sup_{s,t\in[0,T]} X_{p,q}(s,t) > 2u.$$

Let

$$\sigma^2_{p,q} = \sup_{s,t\in[0,T]} \operatorname{Var} X_{p,q}(s,t).$$

We claim that $\sigma^2_{p,q} < 8\ell_*$. Since $\operatorname{Var} D_{ij}(s) \le 2\ell_*$ and $\operatorname{Var} D_{kl}(t) \le 2\ell_*$

$$\operatorname{Var} X_{p,q}(s,t) \le \left(\sqrt{\operatorname{Var} D_{ij}(s)} + \sqrt{\operatorname{Var} D_{kl}(t)}\right)^2 \le 8\ell_*.$$

The first inequality follows from the Cauchy–Schwarz inequality. Equality in the final bound would require both marginal variances to be maximal and equality in Cauchy's inequality. Since p and q are dominant, the marginal variances are maximal only at $s = t = T$. Equality in Cauchy's inequality would then force $D_{ij}(T)$ and $D_{kl}(T)$ to be positively linearly dependent; their variances are both $2\ell_*$, so this means $D_{ij}(T) = D_{kl}(T)$ almost surely. Writing the terminal values as linear combinations of the independent edge increments, the coefficient of the terminal edge of leaf i in $D_{ij}(T)$ is $+1$. This coefficient can be matched in $D_{kl}(T)$ only if $k = i$. Similarly, the coefficient -1 on the terminal edge of leaf j forces $l = j$. Hence $p = q$, a contradiction. Thus $\sigma^2_{p,q} < 8\ell_*$.
The field $X_{p,q}$ has continuous paths and satisfies

$$\mathbb{E}\big\{(X_{p,q}(s,t) - X_{p,q}(s',t'))^2\big\} \le C\big(|s-s'| + |t-t'|\big),$$

so Lemma A.2 yields constants $C < \infty$ and $\alpha < \infty$ such that

$$\mathbb{P}\{E_{ij}(u) \cap E_{kl}(u)\} \le Cu^\alpha \exp\left(-\frac{(2u)^2}{2\sigma^2_{p,q}}\right) = o(\chi_*(u)),$$

because $\sigma^2_{p,q} < 8\ell_*$. Bonferroni's inequalities now yield the asserted union asymptotic. □

Alternative proof of the path asymptotic in Lemma 3.12. Fix $\kappa > 0$, let W_0 be the common Brownian trunk, and put $r = 1 - \kappa/u^2$. Write ϕ_r for the density of $N(0, r)$, and condition on the trunk endpoint $X = W_0(r) = u + z/u$. The reversed, rescaled trunk

$$Y_u(s) = u(W_0(r - s/u^2) - X), \qquad 0 \le s \le n := u^2 r,$$

is a Brownian bridge with mean $-(u^2 + z)s/n$ and covariance $\min(s, t) - st/n$. Its local limit is $W(s) - s$. More precisely, if $a = u^2 + z$, the bridge time change yields

$$Y_u\left(\frac{nt}{n+t}\right) \stackrel{d}{=} \frac{n}{n+t}\left(W(t) - \frac{a}{n}t\right), \qquad t \ge 0.$$

The exponential-martingale hitting formula for Brownian motion with negative drift therefore yields, for $m \ge 0$ and $a + m > 0$

$$\mathbb{P}\left\{\sup_{0\le s\le n} Y_u(s) > m \mid X = u + z/u\right\} = \exp\left(-\frac{2m(u^2 + z + m)}{u^2 r}\right).$$

Thus the full reversed-trunk maximum converges to $E = \sup_{s\ge 0}(W(s) - s)$, where $\mathbb{P}\{E > y\} = \exp(-2y)$ for $y \ge 0$. The conditional bridge is independent of the terminal branches, so E is independent of M_κ in the limit.

Let $q_u(z)$ be the conditional probability of simultaneous crossing at some time in $[0, 1]$. For $z \ge 0$, $q_u(z) = 1$, up to an irrelevant endpoint value. For $y > 0$, the preceding hitting formula and conditional independence yield the exact expression

$$q_u(-y) = \exp(-2y/r) + (1 - \exp(-2y/r))h_\kappa(y), \qquad h_\kappa(y) = \mathbb{P}\{M_\kappa > y\}.$$

Consequently, for fixed z

$$q_u(z) \longrightarrow \mathbb{P}\{z + \max(E, M_\kappa) > 0\}, \qquad \frac{\phi_r(u + z/u)}{u\overline{\Phi}(u)} \longrightarrow \exp(-\kappa/2 - z).$$

For all sufficiently large u, the product of the density ratio and $q_u(z)$ is bounded by $C\exp(-z)$ on $z \ge 0$, and at $z = -y < 0$ by

$$C(\exp(-y) + \exp(2y)h_\kappa(y)).$$

The Gaussian bound on h_κ from the proof of Lemma 3.12 makes this integrable. Dominated convergence and Tonelli's theorem now yield

$$\begin{aligned}\frac{\Psi_\kappa(u)}{\overline{\Phi}(u)} &\longrightarrow \exp(-\kappa/2)\int_{\mathbb{R}} \exp(-z)\mathbb{P}\{z + \max(E, M_\kappa) > 0\}\,\mathrm{d}z \\ &= \exp(-\kappa/2)\mathbb{E}\{\exp(\max(E, M_\kappa))\} = 2\exp(-\kappa/2)\mathbb{E}\{\cosh(M_\kappa)\}.\end{aligned}$$

The final equality follows by conditioning on $M_\kappa = m \ge 0$: $\mathbb{E}\{\exp(\max(E, m))\} = \exp(m) + \exp(-m)$. This representation identifies the separate contributions of the reversed trunk and the terminal genealogy to the local crossing event. □

References

[1] K. Dębicki, E. Hashorva, and L. Wang. Extremes of vector-valued Gaussian processes. *Stochastic Processes and their Applications*, 130:5802–5837, 2020. doi: 10.1016/j.spa.2020.04.008.

[2] K. Dębicki, P. Ievlev, and N. Kriukov. Extremes of Brownian decision trees. arXiv:2512.05248v1, 2025. URL https://arxiv.org/abs/2512.05248v1.

[3] E. Hashorva. Asymptotics and bounds for multivariate Gaussian tails. *Journal of Theoretical Probability*, 18:79–97, 2005.

[4] D. Korshunov and L. Wang. Tail asymptotics for Shepp-statistics of Brownian motion in $\mathbb{R}^d$. *Extremes*, 23:35–54, 2020. doi: 10.1007/s10687-019-00357-z.

[5] V. I. Piterbarg. *Asymptotic Methods in the Theory of Gaussian Processes and Fields*, volume 148 of *Translations of Mathematical Monographs*. American Mathematical Society, Providence, RI, 1996.

[6] L. D. Pitt. Positively correlated normal variables are associated. *The Annals of Probability*, 10:496–499, 1982. doi: 10.1214/aop/1176993872.

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